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Overview of Stark Industries Launch Systems



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Abstract

The present work details the design of the Stark-100k, a single-segment BAllistic Test Evaluation System (BATES) solid rocket motor sized to deliver 100 kN of average thrust at sea level, with a total impulse of 2.5 MN·s and a maximum expected operating pressure of 70 bar. The propellant is a non-aluminized 80/20 AP/HTPB composition, whose ballistic parameters were characterized through small-scale BATES firings and linear regression of the Vieille law. Grain geometry was determined by imposing neutrality, mass-balance, and total-impulse constraints simultaneously. Motor performance was then verified through a fourth-order Runge–Kutta integration of the chamber pressure dynamics equation, with each substep calling NASA CEA in frozen-flow mode and a purposely made discrete grain regression solver. The unsteady simulation confirms satisfaction of all design targets: 100.02 kN average thrust, 2.501 MN·s total impulse, and 69.96 bar peak pressure. A quasi-steady solver, validated against the unsteady model to within 1%, was adopted as the baseline integrator for a Monte Carlo uncertainty campaign propagating variability in a and n. A Rao bell nozzle was designed as well.
Thermal protection is provided by ablative liners in the combustion chamber and in the initial sections of the nozzle, as well as by a multilayer thermal barrier coating at the throat, composed of a bond coat, a YSZ intermediate layer, and an RSZ outer layer, whose combined thickness limits the literature-demonstrated surface temperature within the demonstrated thermal-shock survival envelope. A steady-state heat-transfer model based on Bartz convection and a multi-layer panel method was developed to characterize the heat flux in the throat region, spanning from the $A/A_t = 2$ subsonic section to the $A/A_t = 2$ supersonic section. A cooling jacket with rectangular water channels was dimensioned around this section, and a nitrogen-pressurized blowdown hydraulic system, including tank, valving, distributor and collector manifolds, and NTC thermistor instrumentation, was sized to maintain the coolant below its boiling point throughout the 25-second burn, and to continue operating for 2s before and after firing of the engine.

Acronyms

TBC Thermal Barrier Coating

TBCs Thermal Barrier Coatings

CTE Coefficient of Thermal Expansion

YSZ Yttria-Stabilized Zirconia

BC Bond Coat

TC Top Coat

RSZ quaternary Rare-earth-Stabilized cubic Zirconia

AP Ammonium Perchlorate

HTPB Hydroxyl-Terminated-Poly-Butadiene

MEOP Maximum Expected Operating Pressure

NASA CEA NASA Chemical Equilibrium with Applications

BATES BAllistic Test Evaluation System

LZO Lanthanum Zirconate

DCL Double-Ceramic-Layer

TGO Thermally Grown Oxide

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Chapter 1: Introduction

The advancement of modern solid rocket motor technology requires a rigorous understanding of the coupling between internal ballistics, high-temperature material science and cooling systems. As Stark Industries expands its launch system portfolio in order to sustain the ever-growing demand of SRMs of the Kerbal Space Program, the development of the Stark-100k serves as a critical testbed for characterizing cooling systems' performance in high-thrust motors.

Together with the design of the motor itself, a primary challenge in the design of the Stark-100k is the management of the extreme thermal environment nearby the nozzle throat. To ensure structural safety of the test engine, this project explores the usage of Thermal Barrier Coatings (TBCs) and active cooling methods.

This report is structured to guide the reader through the full preliminary design cycle:

- A technology overview is first presented, covering BATES motors and possible thermal barrier coating layers.
- Ballistic Design: Propellant characteristics are first retrieved from experimental data. The motor BATES geometry is then designed.
- Performance Verification: Utilizing an unsteady pressure solver, integrating NASA CEA with a custom geometrical solver performance requirements satisfaction is verified.
- Thermal Protections: the need for thermal protections is verified, a multi-layered TBC and an ablative system are designed to cope with the intense heat flux in differents parts of the motor.
- Cooling Jacket design: a cooling jacket system is designed, its channels are modeled and sized to quantify expected pressure losses and lay the ground work for the successive support apparatus sizing
- External apparatus sizing: the apparatus required for ground testing is sized, and a specific sensor is selected to conduct the test.

Chapter 2: Technology overview

2.1 Thermal barrier coatings

The primary objective of any thermal protection system is to prevent heat-induced failure of a component, such as the combustion chamber and/or the nozzle [11]. Generally speaking, TBCs reduce the risk of damage to the engine casing [8], moreover, during operations, the lightweight structural components used in rocket engines are subjected to severe thermal gradients and thermal shocks from gases at temperatures reaching 3000 K and pressures exceeding 5 MPa [7], therefore they can be protected using TBCs.

2.1.1 Structure, materials and Properties

When selecting materials for TBCs, several key properties must be considered to ensure optimal performance. The most crucial factors are thermal conductivity, fracture toughness and thermal expansion coefficient[7]. To optimize the TBCs's overall performance, the material should also exhibit a Coefficient of Thermal Expansion (CTE) α similar to that of the metallic substrate, which helps mitigate thermal stresses and prevents delamination. Strong adherence to the substrate is also critical to maintain the integrity of the coating under extreme conditions [14]. Some materials that can be used are: Yttria-Stabilized Zirconia (YSZ), Lanthanum Zirconate (LZO) and rare earth oxides [10]. When TBCs are used in combination with a liquid coolant, the temperature range of the coolant impresses an extreme temperature gradient, inducing large thermal mechanical stresses on the coating [15]. TBCs systems for rocket applications are characterized by a multilayered architecture:

The Bond Coat (BC) Layer is a metallic MCrAlY-type alloy layer (M = Ni, Co, or NiCo) that provides the mechanical bond between the substrate and the ceramic top coat, while its intermediate coefficient of thermal expansion helps reduce thermal mismatch stresses during thermal cycling [10].

The Top Coat (TC) Layer: It is directly exposed to high-temperature combustion gases. In rocket applications it is typically composed of two sub-layers, therefore it's called Double-Ceramic-Layer (DCL): the inner layer is usually made of YSZ (which degrades and progressively loses its properties above about 1200°C), while the outer one can be made of different materials. The YSZ layer is characterized by low thermal conductivity and a porous (oxygen-permeable) microstructure, which enhances its thermal resistance. YSZ is inherently brittle, therefore susceptible to crack initiation and propagation. It is important to keep an YSZ layer before the outer layer for two reasons: it prevents harmful interactions between the outer layer and the Thermally Grown Oxide (TGO), and it improves thermomechanical compatibility. The outermost layer can be made of different materials, in particular, according to [7], the following have been tested: LZO ($La_2Zr_2O_7$), gadolinium zirconate (GZO) ($Gd_2Zr_2O_7$), gadolinium tantalate (GTO) ($GdTaO_4$), and a novel quaternary Rare-earth-Stabilized cubic Zirconia (RSZ) material, composed of 10.6 mol% (Y_2O_3, Yb_2O_3, Gd_2O_3, and Nd_2O_3)-stabilized cubic ZrO_2; this composition stabilizes zirconia in the cubic phase. Among the materials, RSZ shows the best overall match with the requirements of rocket engine applications, with different advantages. RSZ exhibits the lowest thermal conductivity among the considered materials,

approximately 0.60 W/(m·K). Morover RSZ has a coefficient of thermal expansion CTE of $10.1 \times 10^{-6}\,\mathrm{K}^{-1}$, which is very close to that of YSZ ($10.7 \times 10^{-6}\,\mathrm{K}^{-1}$), significantly reducing the risk of premature spallation due to thermomechanical mismatch with both the underlying YSZ layer and the bond coat. In addition, RSZ has a fracture toughness of approximately $1.5\,\mathrm{MPa} \cdot \mathrm{m}^{1/2}$, which, although lower than that of YSZ, remains relatively high among the candidate materials. By combining these properties, RSZ does not necessarily outperform the other materials in a single parameter, but provides the best overall balance of thermal and mechanical performance [7].

The TGO layer is an intermediate layer between the BC and the TC, formed by oxygen diffusion through the porous TC reacting with aluminum in the bond coat. In rocket engines, high temperatures can lead to unstable growth, generating internal stresses that may cause coating delamination [10].

This multilayer configuration (YSZ:RSZ = 1:1, with a total thickness of 350 μm) results in a system capable of withstanding up to 15 thermal shock cycles at 1800°C, a performance level that would not be achievable with a conventional single-layer YSZ topcoat system [7].

2.1.2 Failure mechanisms

The dominant failure mechanisms in TBCs are primarily induced by interfacial delamination and surface cracking [10]. Crack propagation into the substrate can result in critical damage to the cooling channels and even failure of the entire rocket engine. Under tensile loading conditions, a surface crack or a vertical crack formed in one of the layers may deflect and propagate along the interface with another layer or with the substrate [14]. Different modes can be identified for failure of TBCs: in mode A the deflection of the vertical surface crack along the interface of TC and BC occurs and the propagation continues until TC spalls off. In mode B: the penetration of surface cracks into BC occurs, followed by the deflection and propagation along the interface of BC and the substrate, without debonding at the interface of TC and BC. In mode C: this mode involves simultaneous delamination at both the TC-BC and BC-substrate interfaces. These modes are shown in Figure 2.1[10].

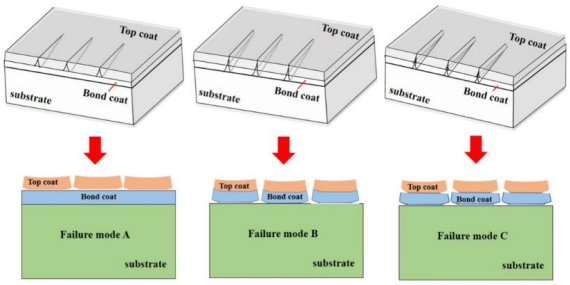


Figure 2.1: Failure modes

In the development of advanced TBCs, researchers conducted extensive testing on various combinations of materials and architectures to overcome the inherent limitations of standard solutions. The possibilities of failure were analyzed with thermal cycling tests in which they compared the thermal cycling lifetime of different coatings with varying thicknesses and layer ratios.

[7].

2.1.3 Manufacturing

Several technologies exist for the application of TBCs; however, this paragraph focuses on plasma spray and on how the characteristics of TBCs vary with changes in the gas mixture used. Plasma spray is a process in which a high-temperature plasma jet melts material particles and propels them onto a surface, where they solidify to form a protective coating, commonly used for thermal barrier coatings TBCs [4].
Argon and Nitrogen plasma gas analysis [27]: Among the plasma gas mixtures, this section focuses on (Ar/H_2) and (N_2/H_2), used to optimize cost and reproducibility while maintaining high performance. This study investigates 8 wt.% yttria-stabilized zirconia (8YSZ) hollow spherical powder (45–75 μm) processed under Ar/H_2 and N_2/H_2 plasma conditions. Nitrogen-based plasmas reach higher particle temperatures (up to 3100°C) compared to argon-based ones (2800°C), while argon generally produces higher particle velocities at a given temperature: deposition efficiency (DE) increases with particle temperature, and is therefore approximately twice as high for nitrogen than for argon. Note that deposition efficiency also represents system productivity. Regarding coating properties, higher temperatures lead to reduced porosity and increased hardness for both gases. Overall, nitrogen results in coatings with lower porosity and higher hardness, whereas argon tends to produce slightly more porous and softer coatings.

2.2 BATES motors

Originally, BATES was born as a system to obtain standardized, reproducible, and accurate ballistic data on solid rocket propellants.
The need for this system arose from the necessity to meaningfully compare the results from different testing facilities. Moreover, it was essential to characterize propellants considering the complex coupling between fluid dynamics, thermochemistry, and structural effects found in real rocket environments which are not captured by a conventional strand burner test [6]. An ideal testing motor shall be characterized by [2]:

- Short duration, to minimize heat loss and nozzle erosion.
- Large throat size to minimize nozzle erosion effects.
- Large port to throat ratio, to minimize grain erosive burning.
- Little or no insulation and liner to avoid unwanted mass contributions to the flow.
- Neutral thrust-time trace at 1000 psia chamber pressure to minimize thrust coefficient corrections and tailoff.
- Nominal expansion to ambient pressure at operating conditions (slightly below optimum area ratio is preferred to guarantee full nozzle flow attachment throughout the firing time).
- Shockless reproducible ignition to ensure proper data acquisition systems and minimum igniter contribution.
- Cylindrically perforated grain configuration to eliminate geometry effects.
- High propellant weight to minimize scale factors with larger motors; low cost per motor, minimum shipping costs, and ease of handling were included as desirable features of a standard motor.

The combination of these parameters led to the development of the modern BATES motor configuration. Its primary characteristic is the nearly neutral burning surface area, resulting in an approximately constant mass flow rate and, consequently, stable chamber pressure and thrust. The dimensions of the first BATES motor were: chamber pressure

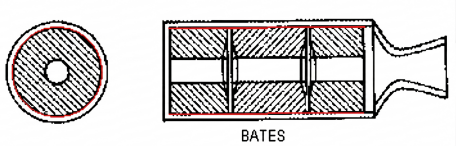
1000 psi, Nozzle Diameter 1 in, Grain Length 12 in, Grain Bore 4.6 in, Grain Diameter 7 in.

2.2.1 Internal geometry and neutrality condition

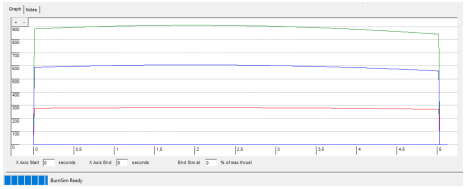
The internal grain geometry consists of a hollow cylinder. The active burning area is composed of the inner cylindrical surface and the two end faces leading to both a radial and axial regression. This configuration ensures that, although the internal radius increases during combustion, the overall grain length decreases correspondingly, maintaining an almost constant total burning surface area.
In Figure 2.2a a sketch of the internal geometry is represented: the red line indicates inhibited (non-burning) surfaces. A notable feature observable in the figure is the presence of grain segmentation through the use of dividers. In larger motors, it is common practice to avoid a single monolithic propellant grain and instead using multiple smaller segments positioned sequentially. If the grain were cast as a single block, the inner burning surface would become much larger than the contribution from the end faces, resulting in a non-neutral propellant grain.
In general the hollow cylinder with burning ends geometry does not guarantee neutrality, to achieve this it is necessary to compute the optimal grain length that ensures constant burning area. From the calculation in appendix 8.6 this length is defined as:

$$L = \frac{3}{2}D + \frac{d_0}{2} \tag{2.1}$$

It is to be noted that this computation ensures only that the initial and final burning surfaces are equal; however, it does not guarantee the absolute neutrality of the grain. In fact deriving the equation for the burn area with respect of the burned thickness gives an expression that in general is not 0. Consequently, the burn area is not perfectly neutral, but has a slightly parabolic behavior. Nevertheless, this effect can be reduced significantly by properly tuning the grain geometric parameters. The near-neutral behavior of the grain is clearly illustrated in Figure 2.2b, in which the temporal evolution of the Klemmung (red), the chamber pressure (green) and the thrust (blue) of a typical BATES motor are represented. A notable aspect emerging from the image is the rapid ignition and extinction of the grain, with practically no tailoff phase. This behavior is a consequence of the cylindrical geometry, which burns uniformly without leaving any propellant trace that can extend the extinguishing times. The simulated grain geometry is the one specified by the parameters given above. The thrust coefficient was selected as $C_T = 1.5$ and the propellant is composed of a single segment. The simulation was performed using BurnSim (version 3.3.10).



(a) BATES grain geometry



(b) Klemmung, pressure and thrust evolution in a 70-pounds BATES motor

Figure 2.2: Grain geometry and typical performance of BATES motor

2.2.2 Propellant parameters from BATES

The first information that can be extracted from a BATES test is the burning rate (r) of a propellant. Thanks to the fact that a single BATES test provides a steady P_c and a known web thickness, the average burning rate can be easily computed as $\bar{r} = \frac{WebThickness}{BurningTime}$. By iterating this process for different pressures, the ballistic exponent (n) and burn rate coefficient (a) can be easily derived, and consequently, the general Vieille law expression can be recovered. The real characteristic velocity (c^*) and consequently the efficiency η_{c^*} can be also derived from this types of motors (Calculations in appendix 8.7).

2.2.3 BATES advantages

Other than the neutrality considerations that have been made previously, this type of motors have further advantages. The absence of complex grain geometries, such as those found in star-shaped grains, simplifies the manufacturing process, making it both easier to produce and cost-effective. The cylindrical configuration gives sliverless combustion [5], ensuring a more precise evaluation of the burned mass. Moreover, the BATES motor grants a high burning area to volume ratio, outperforming the end-burning grain by enabling higher chamber pressures without requiring excessively small throat areas.

2.2.4 Super BATES variation

The Super BATES variation extends the standardized BATES philosophy to larger motors in order to account more realistically for scale-dependent internal-ballistics effects [3]. In large SRM, two-phase gas–particle interactions, wall heat transfer, erosive-burning behavior, and particle residence time do not scale linearly from small ballistic motors. As a result, larger evaluation motors are required to reproduce the internal-flow, combustion, and nozzle environments more faithfully. This type of motor is based on the BATES concept, using the same principles and structure. The main difference lies in the segmentation of the grain. Having a longer motor composed by a single block and imposing neutrality requires either to increase the external diameter or to increase the port one. Being neither of these actions often possible in sensible way, the grain is segmented in smaller sections. Moreover, grain segmentation is structurally better as it reduces stresses and fractures propagation. The dimensions of this variation of the BATES motor vary largely from the normal configuration. Typical flow rates range from 100 to 500 lb/sec, port diameters from 20 to 30 in, and lengths from 50 to 200 in [3].

Chapter 3: Main Assumptions

The presented work is based on the following assumptions:

Motor design, pressure dynamics simulation and thermochemical properties of the combustion produced gas All gas properties are retrieved from the evaluation of NASA CEA rocket problems. The software's MATLAB implementation serves as the backbone of the employed analysis. Pressure dynamics is simulated via an unsteady model during the design phase. Its corresponding Montecarlo implentation employs a quasi-steady dynamical model, that was validated against the unsteady dynamics model benchmark. After an evaluation of erosive burning presence, grain regression rate is assumed to be globally constant, and throat erosion effects are assumed to be negligible. For the purposes of ballistic analysis, the combustion chamber is assumed cylindrical: ablative combustion chamber protections regression rates are not considered in volume evolution over time. The contribution of the pressure vessel head to the total volume is neglected as well to avoid iterative design loops. Only 2D losses were accounted for during the design phase.

Conjugate Heat Transfer Solver (Panels method) Heat flux is restricted to the wall-normal direction by neglecting axial and tangential gradients. Temperatures and fluxes are evaluated via the 1D Planar Composite Wall approximation across the discretized nozzle contour. Panels are generated by extruding the nozzle contour along local normal vectors computed at the segment extremities but each resulting discrete panel is treated geometrically as a composite planar wall with a constant cross-sectional area along its thickness. All variables are computed under a steady-state assumption. Thermal conductivities are assumed constant across the thickness of each structural layer. The transmitted heat flux is assumed to be fully absorbed by coolant convection at the channel interface. Convective boundary conditions, for hot gas side and coolant side, are derived from empirical literature correlations rather than direct CFD resolution. Convective heat transfer between the cooling jacket and the atmosphere is neglected.

Cooling jacket channels design The N equal rectangular-section cooling channels are radially configured around the nozzle in the local axial abscissa direction. Fins thickness between juxtaposed channels are neglected. The coolant is tap water at Mach $M < 0.3$, granting incompressible constant density $997 kg/m^3$, and stationary, empirically modeled properties (8.9), with only bulk temperature being modeled, out of the boundary layer. The piping material is the same as the nozzle material. The heat transfer problem inside the cooling channel considers the convective transfer from the wall to the fluid via convection, neglecting any other form, such as conduction, radiation inside the pipe, and considering the external wall of each pipe thermally insulating with respect to the external testing environment. Under these assumptions, all the heat coming out of the nozzle has to be absorbed by the water mass flow rate. Stationary heat exchange is assumed, so fluid properties are only functions of the axial coordinate. Given the small dimensions the gravitational potential offset is not considered. Dittus Boelter correlation is used and its validity range is then checked a posteriori.

Structural sizing Structural sizing assumptions are presented in the corresponding section 5.2.3

Chapter 4: Engine design

4.1 Design constraints

The Stark-100k is intended to be a test engine, rather than a flight-ready, weight-optimized and performance-optimized unit. Because of this, rather than optimizing the motor for efficiency or for weight, it was chosen to maximize system simplicity and cost-effectiveness. The design process was guided by the following drivers:

- 100kN average thrust at sea-level conditions
- Adapted nozzle at sea-level condition
- 2.5 MNs of total impulse
- 70 bar Maximum Expected Operating Pressure (MEOP), intended as a target value rather than a pressure ceiling

4.2 Propellant composition

The Stark-100k propellant composition was chosen to be 80% in Ammonium Perchlorate (AP) and 20% in Hydroxyl-Terminated-Poly-Butadiene (HTPB) (mass fraction). This specific composition was chosen among admissible ranges [11], in order to guarantee ease of manufacturing without compromising system efficiency excessively. Using a higher AP mass fraction formulation would have led to higher combustion chamber temperatures but it would have reduced ease of manufacturing. On the other end, choosing an higher HTPB mass fraction would have implied lower combustion chamber temperatures, excessively compromising engine efficiency.

4.3 Propellant ballistic properties estimation

The propellant was characterized using a set of small scale BATES motors. The ballistic dataset (Table 4.1)is composed of $N = 15$ $(\tilde{r_b}, \tilde{p})$ measured values, with each line being the process of a BATES motor using the same regression algorithm and reporting the effective test pressure and the corresponding burning rate. The convention followed to identify the $\tilde{p_i}$ values is in accordance with the Stark Industries Launch Systems standards, such that inner compatibility of the results inside the company shall allow comparison between different engine solutions. The testing at different pressures allows therefore the estimation of the ballistic properties. In particular, the adopted ballistic model for the regression is Vieille/Saint-Robert's empirical law, valid under steady burning assumptions:

$$r_b = a\, p^n \tag{4.1}$$

Under assumption of independence and standard gaussian distribution of the measurement errors, respectively $E_1, E_2, ..., E_n$ i.i.d. and $E_i \sim N(0, \sigma^2)$, the linear regression of the logarithmic values allows therefore to determine the pre-exponential factor a and pressure exponent n sample mean values $\bar{a} = 1.6177 \quad [\mathrm{mm}/(\mathrm{s} \cdot \mathrm{bar}^n)]$ and $\bar{n} = 0.38155 \quad [-]$ and the corresponding sample standard deviations $\Sigma_a = 0.0646 \quad [\mathrm{mm}/(\mathrm{s} \cdot \mathrm{bar}^n)]$ and $\Sigma_n = 0.0106 \quad [-]$. These values are in line with typical data and in particular in line

with AP $r_b(p)$ behavior expected within this pressure range, while for higher pressures AP experiences mesa-burning. $n < 1$ is instead proof of a stable-equilibrium controlled combustion. Following classical linear regression statistical theory, under the assumption of independent and equally performed measurements, the sample standard deviations of the line coefficients can also be computed, along with the coefficient of determination $R^2 = 0.99003$ $[-]$ of the regression. Its value, close to 1, is in accordance with our expectation of finding a linear bond, in particular with positive slope (R^2 >0.9). A qualitative normality analysis on the estimations of the r_i standardized residuals $R_i := \frac{\hat{E}_i}{\hat{\sigma}} = \frac{Y_i - (\hat{\alpha} + \hat{\beta} x_i)}{\hat{\sigma}}$ and QQ-norm plot for Shapiro Wilks normality testing would have brought to the same conclusions, therefore highlighting the already well-known explicability of this propellant in Vieille law terms, and allowing a posteriori verification of the assumptions of the linear regression model.

Table 4.1: Ballistic data

$\tilde{p}_i$ **(bar)**	10.1	9.7	10.3	30.2	31.0	29.8	50.2	51.0	50.3	70.2	69.0	69.8	91.2	89.1	89.0
$\tilde{r}_{b,i}$ **(mm/s)**	4.0	3.8	4.1	5.6	6.0	5.7	7.0	7.2	7.1	8.4	8.3	8.6	8.8	9.0	9.2

4.4 Solid propellant grain design

4.4.1 Grain preliminary design

After determining the propellant composition and Vieille law parameters, NASA Chemical Equilibrium with Applications (NASA CEA) was used to retrieve the relevant thermodynamic properties of the combustion gas mixture. This evaluation required a reference combustion chamber pressure, whose specific value of $p_{\text{design}} = 63.96$ bar resulted from an iterative design process. The initial design was centered around this pressure; all variables discussed in the present paragraph refer to the $p_{\text{chamber}} = p_{\text{design}}$ condition. From a sample NASA CEA rocket problem analysis, the expansion ratio $\varepsilon = A_e/A_t$ required for optimal expansion was obtained and fixed at this stage of the motor development. On this note, optimal-expansion requirement was intended to be satisfied at $p_{\text{chamber}} = p_{\text{design}}$ rather than at $p_{\text{chamber}} = p_{\text{average}}$. The design values of c^* and c_f were also determined through this analysis.
A conical nozzle was then defined as a reference geometry. A Rao nozzle was subsequently designed to match the thrust divergence losses of this reference, yielding a 2D loss factor λ. To ensure a net thrust of 100 kN after accounting for divergence losses, a corrected ideal thrust was computed as

$$T_{\text{design}} = \frac{T_{=100\,\text{kN}}}{\lambda}, \tag{4.2}$$

so that an engine sized for $T_{1\text{D}} = T_{\text{design}}$ produces $T_{2\text{D}} = T_{=100\,\text{kN}}$ once 2D losses are applied. With T_{design} established, the target burn time[1], the design condition mass flow rate and the throat area needed were obtained from

$$t_{b_{\text{target}}} = \frac{I_{\text{tot}_{\text{target}}}}{T_{\text{design}}}, \qquad \dot{m} = \frac{T_{\text{design}}}{c_f\, c^*}, \qquad A_t = \frac{\dot{m}\, c^*}{p_{\text{design}}}. \tag{4.3}$$

Once these parameters were fixed, the initial geometry of the Stark-100k BATES grain was computed. Such geometry is uniquely defined by a grain length h_0, an internal

[1] the shown t_{burn} relation is a first approximation and is later corrected through iterative design

hole diameter d_0, and an external grain diameter d_{ext}. To determine these values, three geometrical constraints were imposed, each linked to a desired property of the motor:

Neutrality Condition: The burning surface area $A_b = f(h_0, d_0, d_{\text{ext}}, t)$ at ignition ($t = 0$) was set equal to the burning surface area at the end of the burn ($t = t_b$):

$$A_b(h_0, d_0, d_{\text{ext}}, 0) = A_b(h_0, d_0, d_{\text{ext}}, t_b). \tag{4.4}$$

Mass Balance: An instantaneous mass balance was imposed at $t = 0$ in a form that enforces $\mathrm{d}p_{\text{ch}}/\mathrm{d}t\,(t = 0) = 0$:

$$(\rho_{\text{prop}} - \rho_{\text{ch}})\, A_b(h_0, d_0, d_{\text{ext}}, 0)\, r_b = \frac{p_0\, A_t}{c^*}. \tag{4.5}$$

This formulation serves a dual purpose: it imposes a pressure-stabilising, neutrality-suggesting constraint on the motor, while ensuring that the mass flow rate required to sustain the desired thrust level is indeed produced by the combustion chamber at ignition.

Total Mass: This equation ensures the motor contains sufficient propellant to meet the mission total-impulse requirement:

$$V(h_0, d_0, d_{\text{ext}}, 0)\, \rho_{\text{prop}} = m_{\text{tot}} = \dot{m}\, t_b. \tag{4.6}$$

Despite the imposition of these constraints, a BATES grain is geometrically incapable of delivering a globally constant pressure and thrust profile. This consideration motivates the choice of initial-condition-based constraints rather than integral ones: the inherent behaviour of BATES motors renders any condition based on the integration of the desired constant-performance properties unable to strictly enforce global performance parameters. Initial-condition performance indices were therefore preferred. Equation (4.6) was imposed to link the available propellant mass, and therefore chemical energy, to the total impulse of the system, acting as an anchor of the motor burn time range.

4.4.2 Performance simulation via combustion chamber dynamics model

After the initial grain design process, the actual performances of the designed system were evaluated, to verify requirements satisfaction.
Specifically, the combustion chamber pressure dynamics was integrated via a fourth order Runge Kutta based numerical integration of the global mass balance equation, expressed in the form

$$\frac{dP_{ch}}{dt} = \left(\frac{V}{\frac{R}{M_w}T}\right)^{-1} [(\rho_p - \rho)A_b r_b - \frac{P_{ch}A_t}{c^*}] \tag{4.7}$$

The derivation of 4.7 and the related underlying hypotheses are expressed in Appendix 8.4, while the NASA CEA-based numerical integration strategy is detailed in appendices 8.4 and 8.5.1. In short, each RK4 substep relies on the solution of a NASA CEA rocket problem to retrieve the intermediate chamber gas thermodynamic properties, coupled with a discrete geometrical solver that simulates grain regression. The obtained thermodynamic and geometric parameters are then fed into 4.7, obtaining a differential equation in the form

$$\dot{P_{ch}} = f(P_{ch}, \hat{x}, t) = f(P_{ch}, \begin{bmatrix} \mathbf{x}_{\text{CEA}} \\ \mathbf{x}_{\text{geometrical}} \end{bmatrix}, t) \tag{4.8}$$

where $\mathbf{x}_{\text{CEA}}, \mathbf{x}_{\text{geometrical}}$ are vectors containing variables retrieved from NASA CEA and the coupled geometrical solver respectively. On this note, given the absence of meaningful erosive burning phenomena, as shown in 4.4.3, the propagation of geometrical properties was based on a coarse yet representative grain cross section discretization.
To ensure methodological consistency, frozen-flow NASA CEA simulations were adopted for both thermal and ballistic analysis. The equilibrium model in NASA CEA can yield unreliable transport properties, which would impair the results of the thermal analysis. A possible workaround would be to run an equilibrium-mixture simulation while substituting frozen-flow transport properties where needed; however, this approach entails mixing equilibrium-based and frozen-based quantities, introducing an internal inconsistency. A lower-fidelity yet self-consistent model (frozen mixture) was therefore preferred over a higher-fidelity but internally inconsistent one (equilibrium composition with frozen-flow transport properties).
The ignition and deflation pressure transients were also simulated. The ignition transient is comprehended in the main motor simulation, thanks to the imposition of an initial condition of $P_{ch}(0) = 10$ bar. This numerical approach implies assuming the presence of an igniter capable of instantly generating a $P_{chamber} = 10$ bar and instantly spreading the flame over the whole initial grain surface.
The deflation transient was modeled with a separate integration routine, consisting in a modified version of the RK4-integrated 4.7 used for the main simulation. Its contribution to motor performance was neglected, most importantly due to the significantly lower fidelity of this simulation, which meaningfully approaches the limits of validity of the 4.7 model, as explained in Appendix 8.5.3. This second integration routine uses the results of the first simulation as initial conditions, and is included to show a qualitative chamber deflation analysis.

4.4.3 Erosive burning analysis

Erosive burning refers to any increase in the propellant burning rate caused by the high-velocity flow of combustion gases across the burning propellant surface. It can seriously affect the performance of solid propellant rocket motors. It occurs primarily in port passages or in grain perforations as the combustion gases flow toward the nozzle; it is more likely to occur when the port passage cross-sectional area A is not large relative to the throat area A_t, with a port-to-throat area ratio of 5 or less. [11]
To verify the presence of meaningful erosive burning effects, the aforementioned ratio was evaluated. The designed grain has an A_0/A_t value of 20.2, which suggests the absence of meaningful erosive burning. The effect of this phenomenon was nonetheless quantified using the model first developed in 1956 by Lenoir and Robillard and successively improved. [11] ($r = r_0 + r_e$) The erosive burning induced contribution to the burning rate r_e was quantified in the most critical grain location via the model expressed in 8.3, and was determined to be $\ll 0.01 \times r_0$, where r_0 is the pressure-induced burning rate, therefore it was deemed negligible in our analysis

4.4.4 Optimization and final geometry

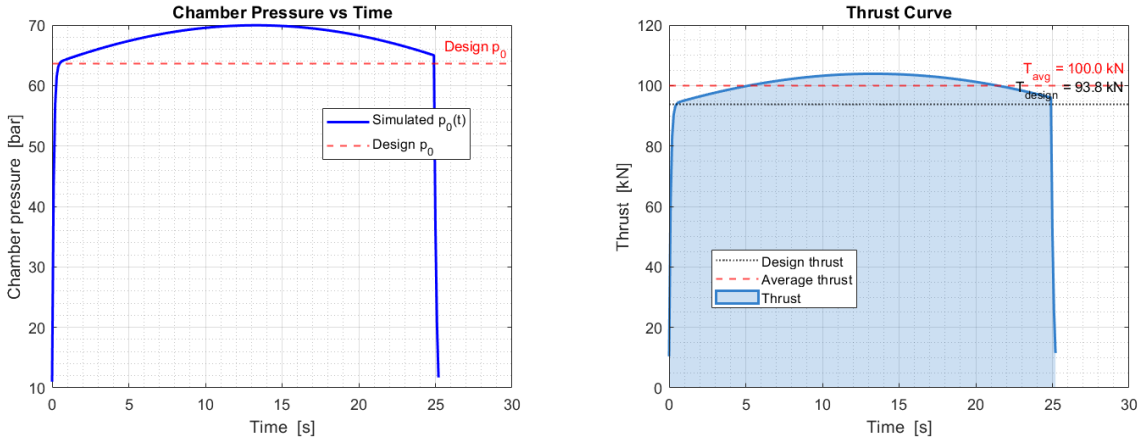
Following the initial design process, the obtained BATES geometry was optimized to meet the given design constraints. The optimization process led to a BATES engine with the following geometrical properties:

DATA	SYMBOL	VALUE	UNIT
grain height	h_0	1632.40	mm
grain internal diameter	d_0	498.89	mm
grain external diameter	d_e	902.91	mm
grain thickness	w	202.01	mm
Init. burning area	$A_{b,0}$	3.44810	m^2
Throat diameter	d_t	111.00	mm

Table 4.2: Motor Grain data

4.4.5 Performance results

Unsteady $P_{chamber}$ model: simulation results The designed motor simulation led to the performance parameters shown in Table 4.3, corresponding to the results shown in figure 4.1



(a) Pressure trace.

(b) Thrust trace.

Figure 4.1: Simulated chamber pressure and thrust traces.

Comparison with quasi-steady p_{ch} model Given the relations

$$p_c = \left(\rho_{\text{prop}} \frac{A_{\text{burn}}}{A_t} a\, c^*\right)^{\frac{1}{1-n}}, \qquad z(t) = \int_0^t p_c(\tau)^n \, d\tau = \int_0^t r_b(\tau)/a \, d\tau \tag{4.9}$$

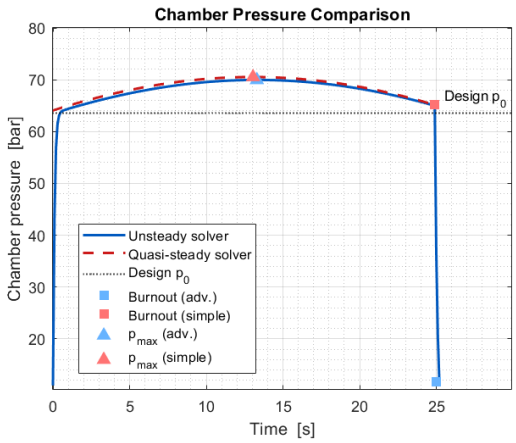
Differentiating and substituting evaluated at $A_b(a\, z(t))$, one obtains:

$$\dot{z} = \left(a\, \rho_p\, C^* \frac{A_b(az)}{A_t}\right)^{\frac{n}{1-n}}, \qquad z(0) = 0 \tag{4.10}$$

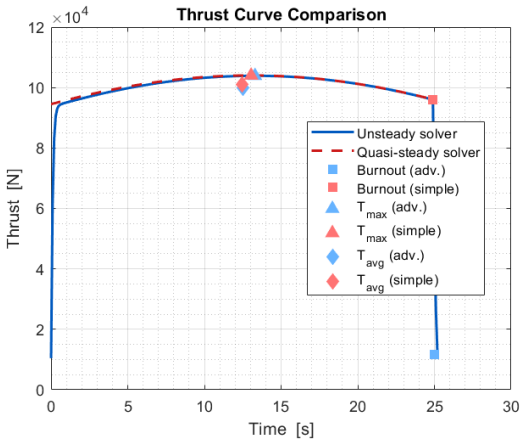
Once $z(t)$ is known, the chamber pressure is recovered as:

$$p_c(t) = [\dot{z}(t)]^{1/n} \tag{4.11}$$

The comparison between the unsteady and quasi-steady models is shown in Figure 4.2.



(a) Pressure comparison.



(b) Thrust comparison.

Figure 4.2: Pressure and thrust curves comparison

Table 4.3 summarises the key performance metrics obtained from both solvers, as discussed in the comparison presented in figure 4.2.

Table 4.3: Simulation comparison summary.

Parameter	**Unsteady solver**	**Quasi-steady solver**
Burn time	25.00 s	24.89 s
Total impulse	2500601.7 Ns	2515146.3 Ns
Average thrust	100024.07 N	101046.42 N
Peak thrust	103885.13 N	104009.42 N
Peak chamber press.	69.960 bar	70.522 bar
Design chamber p.	63.600 bar	63.600 bar
Specific impulse	220.34 s	221.63 s

Critical analysis of simulation results The unsteady simulation yields an average thrust of 100.02 kN and a total impulse of 2.5006 MNs, effectively matching the 100 kN and 2.5 MNs design targets. The peak chamber pressure of 69.96 bar satisfies the 70 bar MEOP constraint. The design pressure $p_{\text{design}} = 63.96$ bar was intentionally chosen to push the peak pressure as close as possible to the MEOP ceiling, thereby maximizing motor performance within the allowed envelope. The quasi-steady solver predicts a peak pressure of 70.52 bar, formally exceeding the MEOP. This discrepancy arises because the quasi-steady model neglects the finite chamber filling dynamics that attenuate the pressure peak. The two models agree to within $\sim$1% on average performance metrics. The close agreement between the two solvers on burn time (25.00 s vs 24.89 s), average thrust, and total impulse confirms that the quasi-steady assumption captures the bulk of the dynamics adequately. The specific impulse results are also comparable: both simulations show an $I_{sp} = I_{tot}/(m_{tot}g_0)$ lower than the CEA computed value (225.54s), likely due to the effect of 2D losses. This conclusion justifies the adoption of the quasi-steady solver as the baseline integrator for the Monte Carlo campaign described in Section 4.5.2, where the computational savings afforded by this model are essential to make a large-sample simulation campaign tractable.

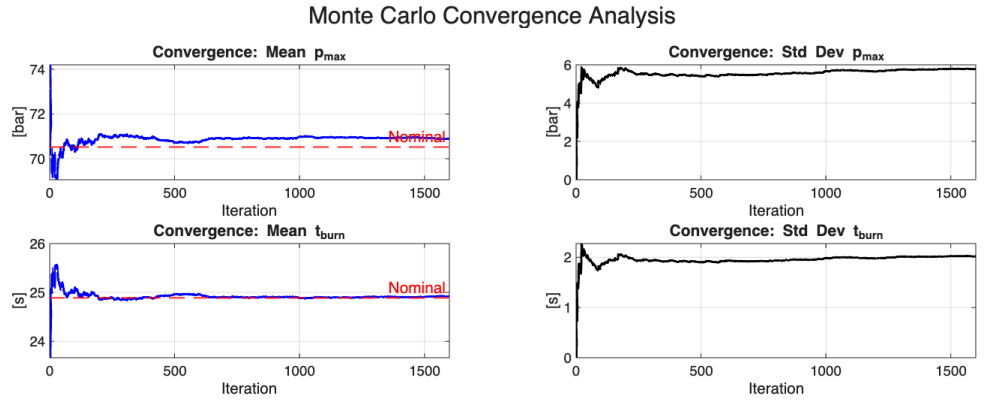
4.5 Uncertainty estimation

4.5.1 Possible uncertainty sources

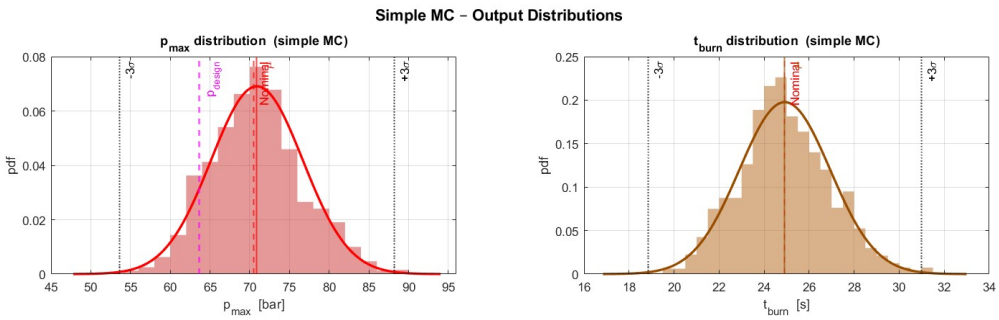
The provided dataset allowed to retrieve the estimated values for the means of a and n, together with their related uncertainties. However, in a real-case scenario, other variables of this problem are subjected to uncertainties and must be treated accordingly. Specifically, manufacturing tolerances affect the real values of $h_0, d_0, d_{\text{ext}}, \rho_{propellant}$ and A_t, and therefore affect the estimated performance parameters.

4.5.2 Monte Carlo campaign

For the case at hand, only Vieille's Law parameters a and n with real mean values μ_A, μ_n where taken into consideration for the uncertainty study. Note that a larger set of variables onto which iterate a Monte Carlo campaign would have led to a far greater computational effort request. A fortiori, in this case only a and n come from a given dataset (as described in 4.3), therefore are suitable for this approach. All other variables are here considered deterministic. For the study of the uncertainties, a Monte Carlo approach was followed. From the linear regression model, $\bar{a}$ and $\bar{n}$, estimators of $E(a)$, $E(n)$, are gaussian are known. Therefore, a set of N=40 gaussian stochastic seeds was planted for both, also because uncertainties on a and n can be led to manufacturing. Later on, an ordered set of a and n seeds coupled was allocated, and ultimately fed into a shuffle function to properly monitor Monte Carlo convergence. The quasi-steady ballistic simulation was then run N^2 times, once for each of the the NxN couples of (a_i, n_i).



(a) Convergence of the means



(b) Estimated distributions

Figure 4.3: Monte Carlo simulation results

Given that N is 'sufficiently high', the estimated sample mean values $\widehat{\overline{T_m}}$, $\widehat{\overline{I_{tot}}}$, $\widehat{\overline{P_{max}}}$, $\widehat{\overline{t_{burn}}}$,

of the four aleatory variables $T_m, I_{tot}, P_{max}, t_{burn}$ converge (Examples for MEOP and burning time shown in 4.3a) to their real mean value $E[T_m] \simeq \widehat{\overline{T_m}} = 101601.2$ N, $E[I_{tot}] \simeq \widehat{\overline{I_{tot}}} = 2515167$ Ns, $E[P_{max}] \simeq \widehat{\overline{P_{max}}} = 70.906$ bar, $E[t_{burn}] \simeq \widehat{\overline{t_{burn}}} = 24.92$s given that the sample mean value is a mean square consistent and asymptotically unbiased estimator of the mean, and given that here we have a sufficiently large random sample N, also in accordance with the Weak Law of Large Numbers (WLLN) and Strong Law of Large Numbers (SLLN). The same can be said for the standard deviations $\sqrt{\text{Var}(T_m)} \simeq \hat{s}_{T_m} = 8252.4$ N, $\sqrt{\text{Var}(I_{tot})} \simeq \hat{s}_{I_{tot}} = 28.333$ Ns, $\sqrt{\text{Var}(P_{max})} \simeq \hat{s}_{P_{max}} = 5.7760$ bar, $\sqrt{\text{Var}(t_{burn}} \simeq \hat{s}_{t_{burn}} = 2.02$s. Additionally, the Central Limit Theorem (CLT) embeds gaussianity of these means (Figure 4.3b). Note that the mean values do not converge of the nominal values computed using the mean values of a and n, due to the strong nonlinearities embedded in the simulation algorithm, in accordance with Jensen Inequality Theorem. In particular, the nominal values underestimate the mean expected values.

4.6 Nozzle design

Upon determination of the area ratio $\epsilon = A_e/A_t = 7.1817$ $A_e = 0.0695$ m^2, $A_t = 0.0097$ m^2, of the nozzle, and having set already $R_{cc} = 0.4515$ m, a conic nozzle divergence angle of $\alpha = 15°$, the 2D divergence losses have been determined and accounted for as $\lambda_{2D} = 0.9830[-]$. For the nozzle geometry, though, an equivalent-2D loss bell nozzle geometry has been chosen. In this way, the previous ballistic design that used the same loss is still consistent with the ideal performance obtainable with this new bell shape nozzle. The isoloss inner Bell shape of the nozzle is computed via Rao approximation [11] [35] [38]. The equivalent Rao bell profile $r_{RAO}(x)$ is therefore found (8.1) with same convergent angle $\beta = 45°$, convergent and divergent lengths of respectively $L_{C,RAO}$=0.4304 m, $L_{D,RAO} = 0.2801$m $= 0.805 \cdot L_{D,CONIC}$ (fig:8.4). The resulting C^1 profile can be therefore differentiated with respect to x, and the expression of the tangent and normal unit vectors at each x on the curve can be computed (8.1). Also, it can be integrated to obtain the curvilinear length $l_{RAO}(x)$of the shape at each x (8.1). Since the analytic expression of the Rao profile is known, the analytic integration can be performed quite easily (8.1). This allows for an exact discretization of the curve along its local curvilinear abscissa into a chosen set of N curvilinear elements all of length dl. All thermodynamic properties computed by NASA CEA are referred to the nozzle x-axis position. In order to track the evolution of any thermodynamic property $\mathcal{P}(x)$ along the nozzle profile, it is therefore possible, by function composition, to compute $\mathcal{P}(l_{RAO}(x))$. Furthermore, given the injectivity and surjectivity of $l_{RAO}(x)$, the function is a bijection, therefore $x(l_{RAO})$ exists, giving the $\tilde{x} = x(\tilde{l}_{RAO})$ position in the nozzle associated to any intermediate length $\tilde{l}_{RAO}$ along the Rao-shaped profile.

4.7 Ablative Thermal Protections design: nozzle and combustion chamber

Two separate ablative materials were considered for both the combustion chamber and the first section of the convergent in order to size the ablative thermal protections: composites with a matrix of phenolic resin and fibers of Rayon carbon fibers. Rayon based Carbon fiber is preferred for ablative applications as it provides lower thermal conductivity and

higher Inter-laminar Shear strength. The second material was chosen as it has lower thermal conductivity coefficient and lower erosion rate [13]. The required thickness of the ablative layer was estimated using the following approximation for surface regression and thermal penetration (See Appendix 8.2):

$$t_{Ablative} = S_f \left(\dot{r} \Delta t_{burn} + \sqrt{\alpha_{\perp} \Delta t_{burn}} \right) \tag{4.12}$$

where $\dot{r}$ is the surface erosion rate (0.029 mm/s), $\alpha_{\perp}$ is the perpendicular thermal diffusivity, and Δt_{burn} is the propellant burn time. A safety factor of $S_f = 1.5$ was applied to the final thickness, the final result is provided by Table 4.4.

Parameter	**Symbol**	**Value**	**Unit**
Burn Time	Δt_{burn}	25.00	s
Erosion Rate	$\dot{r}$	0.029	mm/s
Thermal Diffusivity	$\alpha_{\perp}$	0.0036	cm^2/s
Eroded Thickness	–	0.725	mm
Virgin Insulation	–	3.000	mm
Safety factor	S_f	1.5	[-]
Total Required Thickness	$\mathbf{t_{Ablative}}$	**5.588**	**mm**

Table 4.4: Ablative Thermal Protection System.

The ablative coating is designed for application on the combustion chamber walls (including the injector head) and the first uncooled sections of the nozzle. This section is specifically defined by area ratios where $A_x/A_t < 2$ in the subsonic regime. An independent sizing analysis must be conducted for the uncooled, supersonic divergent section of the nozzle. Because this region undergoes rapid flow expansion, it is subjected to a drastically different thermal load compared to the combustion chamber and the convergent section, requiring the ablative thickness to be specifically tailored to the local heat flux.

4.8 Thermal Barrier Coatings design

The TBC adopted for this solution is a double-layer ceramic system based on:
-a bi-layer top coat, where the outermost layer in contact with the combustion gases is made of RSZ, for which a thermal conductivity of 0.6 W/(m·K) is assumed, and an inner layer made of YSZ, for which a thermal conductivity of 0.9 W/(m·K) is assumed. [7]
-a NiCrAlY bond coat (Ni-17%Cr-6%Al-0.5%Y (V2-03-528)), for which the thermal conductivity is computed using the law given by [12], as a function of the interface temperature between the bond coat and the top coat.
The suitability of the selected TBC for the application considered is justified as follows: according to [7], for a configuration (YSZ:RSZ = 1:1, with a total thickness of 350 μm), the TBC survives at 15 cycles and at a maximum temperature of 1800°C. Given that the selected configuration (YSZ:RSZ =1:1 with a slightly reduced overall thickness) and that the TBC is required to withstand a single cycle of 25 seconds with a maximum temperature of approximately 1670°C, the survival of the TBC has been assumed valid.

4.9 TBC Performance via panel method

In order to estimate the steady state temperature profile along the section of the nozzle, a low fidelity conjugate heat transfer model was employed. For the complete formulation

and implementation of the model refer to 8.8.
The nozzle boundary is discretized into a five-layer domain characterized by varying thicknesses and thermal conductivities. Relative to the primary nozzle contour, two layers project radially outward, comprising the main structural wall and the cooling jacket channel while three thermal barrier layers (bond coat, ceramic inner layer, and ceramic outer layer) project radially inward to form the hot-gas interface. Geometrically, the TBC analysis is restricted to the high-heat-flux throat region: the longitudinal computational domain spans exclusively from a subsonic area ratio of $A_x/A_t = 2$ to a supersonic area ratio of $A_x/A_t = 2$.Figure 4.7 shows the computational domain for the specific zone of interest and the layer structure.
The simulation was conducted by imposing the temperature profile on the inner wall of the cooling jacket pipes, being the outermost wall of the material mesh, $T_{OuterWall}(x)$, being x the axial coordinate of the nozzle to ensure that the water used as coolant does not reach boiling point. Since the water is discharged at a pressure of 20 bar, the boiling point would be reached at a temperature of 485 K. Therefore, a $T_{OuterWall}(x) = 455.15$ K =const. profile was chosen, since this is the temperature of the wall in contact with the water flow, a 30 K margin was imposed to ensure that the coolant never reaches boiling point. In order to obtain a solution, the equivalent free stream temperature profile for the hot gas (adiabatic wall temperature) in the nozzle is derived for every panel according to:

$$T_{AdiabaticWall} = R(M, \gamma) \cdot T_0$$

with correction factor $R(M, \gamma)$, computed as function of the locally sampled mach number M and specific heat ratio γ, and total temperature T_0. Values for M, γ, T_0 are computed by a custom NASA CEA based tool for every panel (see Appendix 8.15) of the mesh and the local profile for $T_{AdiabaticWall}(x)$ is retrieved, under the assumption that the gas properties are not influenced by the heat flux that leaves the nozzle. After an iterative design process, the nozzle layers thickness values were derived with respect to the following constraints:

- the bond coat interface reaches a temperature that for which $K_{bondCoat}$ is consistent with the thermal conductivity value at that temperature $K_{bondCoat} = K_{bondCoat}(T)$ [12] (See Appendix 8.14).
- the thickness for both ceramic layers are equal and within acceptable range for manufacturing [7]
- the outer ceramic layer temperature profile never exceeds 1800 C°.

Table 4.5 reports the characteristics of every layer within the final nozzle mesh:

Layer Component	**Thickness, t (mm)**	**Thermal Conductivity, K (W/m·K)**
Cooling Jacket pipe wall (Copper)	2.00	260
Structural Wall (Copper)	10.00	260
Bond Coat	0.06	18.78
Intermediate Layer	0.12	0.90
Ceramic Top Coat	0.12	0.60

Table 4.5: Geometric and thermophysical properties of the multi-layered nozzle wall.

The defined mesh and boundary conditions are utilized to compute the steady-state thermal solution. The remainder of this section presents and discusses the results of this

analysis. Figure 5.2a illustrates the axial temperature profile at each interface. The plots clearly illustrate the critical role of the (TBC). Despite its minimal thickness, the TBC induces a severe thermal gradient, drastically reducing the wall temperature and guarding the underlying nozzle material.
The analysis demonstrates that the current design of the multi-layer ceramic coating represents a valid engineering solution. The operating temperature of the innermost layer reaches approximately 1940 K (1667 °C), which falls safely within the operational thermal limits established by [7] while copper-based layers temperature reaches approximately 625 K which perfectly falls within the temperature range for the alloy ([20]):

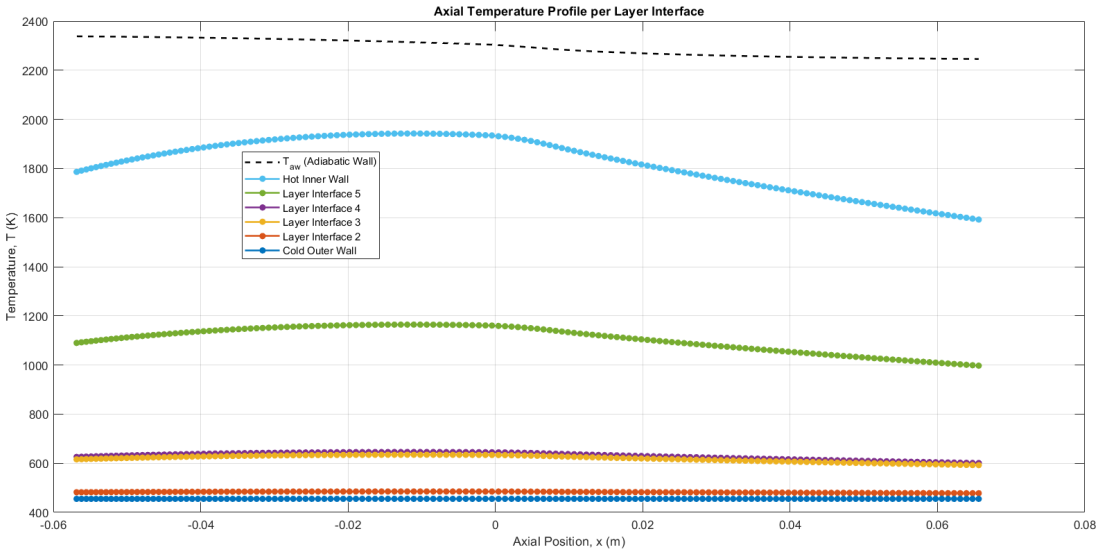


Figure 4.4: Temperature profile along axial nozzle direction.

Another crucial piece of information to interpret the temperature profile is the local heat flux through every panel of the mesh, since the temperatures profile is computed with the assumption that the heat passing for every panel is completely absorbed by the coolant through the cooling jacket in order to obtain a steady state solution.
The convective heat transfer coefficient, h_{HotGas}, is computed according to the Bartz model (See Appendix 8.8), depending on only one unknown $T_{WallGas}$ through the property variation correction factor $\sigma(T_{WallGas})$. This term accounts for the steep temperature and density gradients across the boundary layer (See Appendix 8.8) [19], depending also on the known M, T_0 and γ. Since the boundary conditions for the problem are expressed in terms of $T_{AdiabticWall}$ computed by CEA and $T_{OuterWall}$ as already discussed before, for every panel an iterative process is needed to compute the temperature of the inner wall of the nozzle ($T_{wallGas}$), according to :

$$\dot{q} = \frac{(T_{AdiabaticWall} - T_{OuterWall})}{R_{Tot}(T_{wallGas})} = h_{HotGas}(\sigma(T_{WallGas}))(T_{AdiabaticWall} - T_{wallGas}) \quad (4.13)$$

A more detailed description of implementation is found in Appendix 8.8. Figure 4.5 and Figure 4.6 show the local heat flux and the local heat flux convective coefficient along the axial nozzle direction x: the maximum heat flux value is located slightly before the throat section of the nozzle, as suggested by literature.

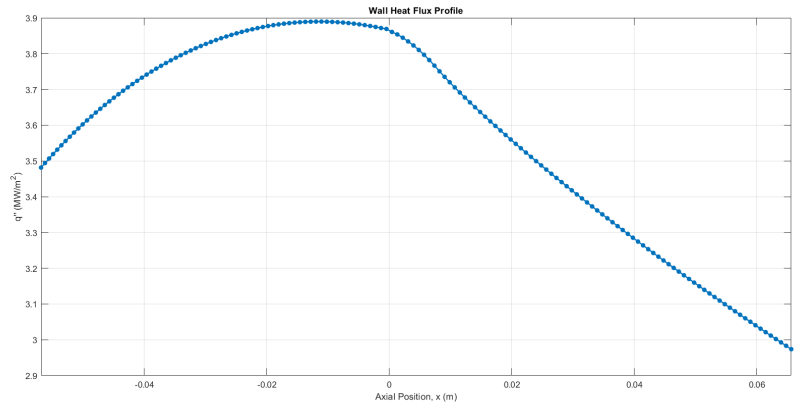


Figure 4.5: Axial distribution of the through-wall heat flux under steady-state conditions (with TBC).

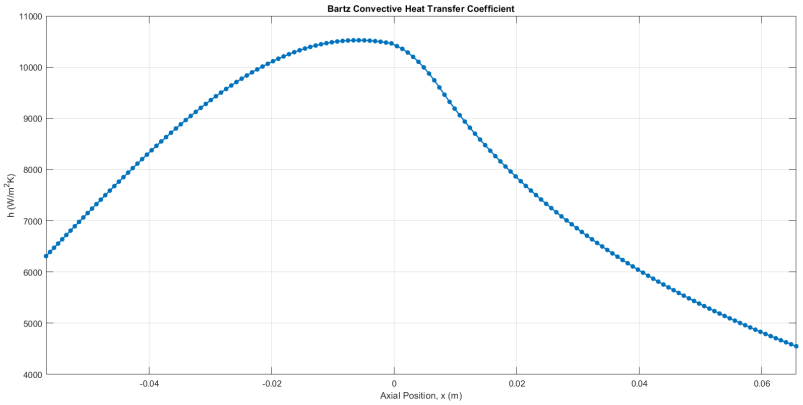


Figure 4.6: Axial profile of the convective heat transfer coefficient (with TBC).

4.9.1 Performance without TBC

To justify the presence of the thermal barrier coatings, a complementary analysis was conducted to assert the system performance without TBCs. Following the same procedure, starting from the same nozzle discretization criterion, only two layers of material (nozzle layer, and lower water channel wall) were grown in the outward direction, with the same specified thicknesses as the previous section to ensure consistency with the previously designed nozzle and cooling channel size. A view of the mesh (for both coated and uncoated design) is show in Figure 4.7 and Figure 4.8 :

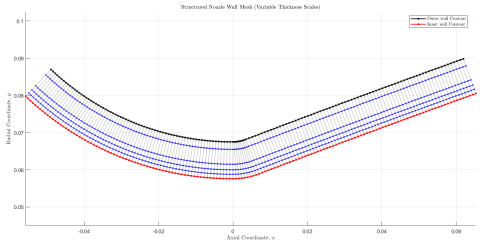


Figure 4.7: Nozzle subMesh with TBC (thickness exaggerated).

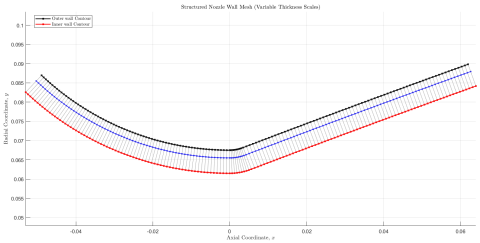


Figure 4.8: Nozzle subMesh with no TBC (thickness exaggerated).

The same external boundary condition ($T_{OuterWall}$) that was chosen considering the water properties at $\approx 20\,\text{bar}$ was enforced on the outer boundary and both temperature and heat flux profiles were computed. Figure 4.9 show axial temperature distribution plotted along the nozzle axial direction:

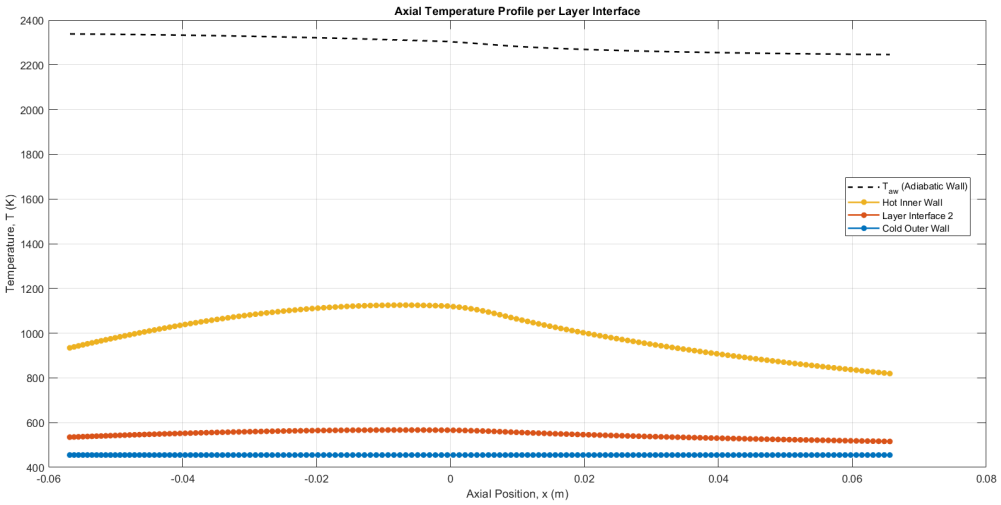


Figure 4.9: Temperature profile along axial nozzle direction.

As expected, the operating temperature of the nozzle, which is in direct contact with the hot combustion gases, increases significantly from approximately 625 K to 1124 K. This thermal excursion would severely degrade the mechanical properties of the GRCop-84 alloy [20], ultimately leading to nozzle failure. Beyond risking structural failure, the absence of the TBC drastically amplifies the thermal loads transferred to the coolant. The resulting heat flux and convective heat transfer coefficient are detailed in Figures 4.10 and 4.11:

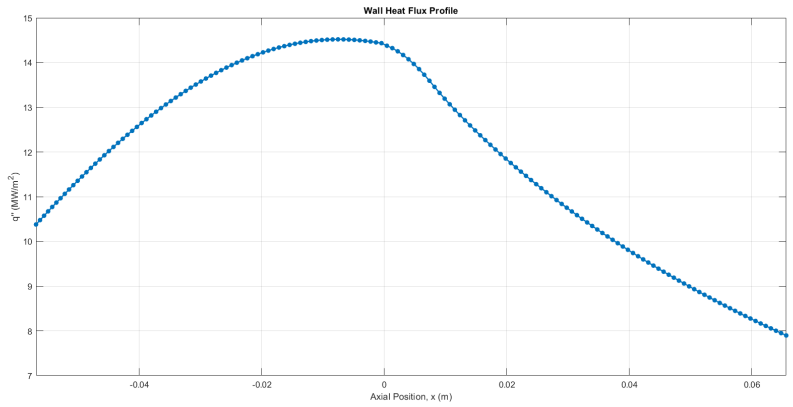


Figure 4.10: Heat flux profile along axial nozzle direction

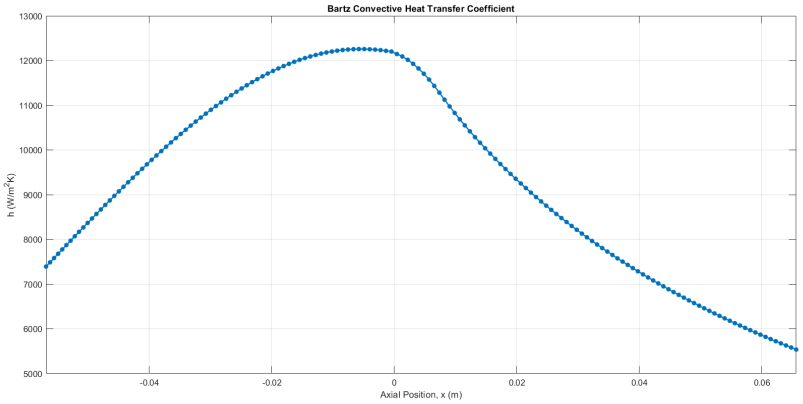


Figure 4.11: Heat flux coefficient profile along axial nozzle direction

Since the cooling jacket is expected to be designed to provide sufficient convective heat flux, which along one single panel is computed as

$$\dot{q}_{Panel} = \frac{T_{AdiabaticWall} - T_{OuterWall}}{R_{Tot}}, \qquad R_{Tot} = \frac{1}{h_{HotGas}(T_{wallGas})} + \sum_{i=1}^{N_{layers}} \frac{t_i}{k_i} \tag{4.14}$$

Assuming the outer wall temperature ($T_{OuterWall}$) is held constant by the cooling jacket, the total heat flux across a given panel ($\dot{q}_{Panel}$) is governed entirely by the total thermal resistance (R_{Tot}). As illustrated in Figure 4.6, the peak convective heat transfer coefficients (h_{HotGas}) remain of the same magnitude for both configurations: $\approx 12000\,\mathrm{W/(m^2 \cdot K)}$ with the TBC and $\approx 10000\,\mathrm{W/(m^2 \cdot K)}$ without. Consequently, the convective resistance term ($1/h_{HotGas}$) is comparable in both cases. Instead, R_{Tot} is dominated by the conductive summation term ($\sum t_i/k_i$). Since the thermal resistance of the highly insulative TBC layer (t_{TBC}/k_{TBC}) is an order of magnitude greater than that of the copper substrate, omitting the coating drastically diminishes R_{Tot}. As confirmed by the resulting plots, this reduction forces a massive increase in the overall heat flux absorbed by the uncoated nozzle panels.

Chapter 5: Cooling jacket design

5.1 Design of the cooling jacket

The cooling jacket consists of a series of channels routed along the exterior contour of the nozzle in the throat region from $A(x)/A_t = 2$ in the convergent to $A(x)/A_t = 2$ in the divergent. As the coolant flows through these passages, it absorbs thermal energy from the nozzle wall and through the piping via forced convection, resulting in a sensible heat rise within the fluid.

5.1.1 Model and implementation

A heat transfer problem is solved for each element of length l of the external solid boundary of the mesh, starting from the element closest to the combustion chamber, for which the initial water temperature is known and imposed as a boundary condition. Considering the whole jacket, each panel is therefore a heat problem with a known initial inlet water temperature T_i in x_i and an unknown final water T_f in x_f, with a prescribed heat flux per unit surface $q = Q/S$, where Q is the heat flux though the conical segment, computed via convective heat exchange at the TBC-gas interface and S is the conic surface covered, from x_i to x_f. The panel wall temperature is known as it is the same profile temperature imposed on the outer boundary of the mesh ($T_{OuterWall(x)}$). $c_p(T)$, $K(T)$, $\mu(T)$ are functions of T. Firstly, $T_f = T(x_f)$ is computed, and then also the convective heat transfer coefficient h is evaluated

$$T_{\mathrm{f}} = T_{\mathrm{i}} + \frac{q_{\mathrm{vec}}}{\dot{m}/S_{vec} \cdot c_p(T_{\mathrm{i}})} \quad , \quad h_{\mathrm{local}} = \frac{q_{\mathrm{vec}}}{T_{\mathrm{panel}} - T_{\mathrm{f}}} \tag{5.1}$$

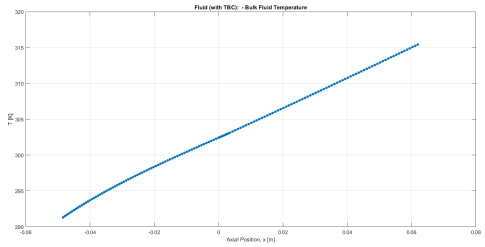
A system of 3 equations in three unknowns (b, v_f, D_h), channel height, water speed in x_f, and hydraulic diameter is then solved, embedding Dittus Boelter Nusselt correlation, the section geometry and mass flow rate conservation.

$$\mathbf{F} = \begin{bmatrix} 1 - \frac{\dot{m}(l,b,v_f)\cdot N}{\dot{m}} \\ 1 - \frac{\mathrm{Nu}(h_{local},D(a,b),T_{\mathrm{f}})}{\mathrm{Nu_{DB}}(\mathrm{Re}(T_f,v_f,D(a,b),\mathrm{Pr}(T_f))} \\ 1 - \frac{D_h(a,b)}{D} \end{bmatrix} = \mathbf{0} \tag{5.2}$$

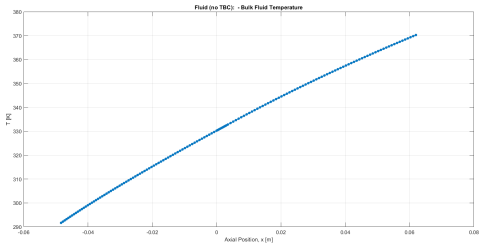
5.1.2 Cooling jacket comparison

This section discusses the steady state thermal analysis results and performance characteristics of the cooling jacket, comparing configurations with and without the TBCs. To facilitate a direct assessment of performance and physical realizability of the channels, both simulations share an identical boundary condition, specifically enforcing the same prescribed temperature profile on the outer wall (455.15 K), mass flow rate of 2.0 $\frac{\mathrm{Kg}}{\mathrm{s}}$ of water (assumed to be evenly distributed along all channels) and an initial inlet temperature for the water of 291.15 K.
Figure 5.1 compares the bulk fluid temperature profiles for both configurations. As discussed in Section 4.9.1, the absence of the TBC drastically reduces the overall thermal resistance of the wall, leading to a substantially higher heat flux across the walls. Since the outer wall temperature profile is strictly imposed as a fixed boundary condition, the

cooling fluid must absorb this increased thermal load. Consequently, the cooling jacket without the thermal coating exhibits a significantly higher bulk fluid temperature distribution. The uncoated system exhibits a $\Delta T_{NoTBC} \approx 89\,\mathrm{K}$ compared to the coated design, with $\Delta T_{TBC} \approx 24\,\mathrm{K}$.



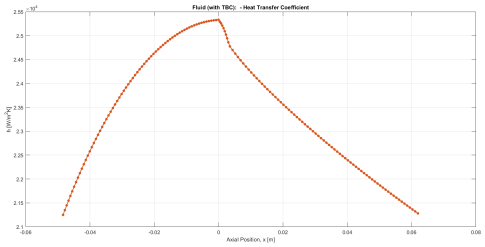
(a) Temperature profile along axial nozzle direction with TBC.



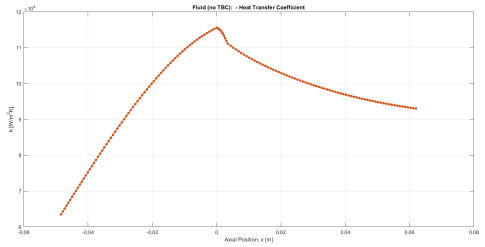
(b) Temperature profile along axial nozzle direction with No TBC.

Figure 5.1: Bulk fluid temperature profile comparison.

To satisfy the steady-state assumption, the cooling jacket without the TBC must exhibit a significantly higher convective heat transfer coefficient to dissipate the increased thermal load, as illustrated in Figure 5.2:5.2. The plots shows a convective heat transfer coefficient h_{NoTBC} for the uncoated design of approximately $115550 \frac{\mathrm{W}}{\mathrm{m^2K}}$, while the coated design reaches the lower value of h_{TBC} of approximately $25333\,\frac{\mathrm{W}}{\mathrm{m^2K}}$.



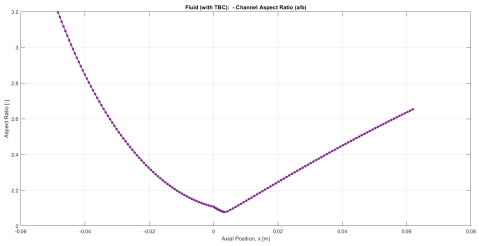
(a) heat transfer coef profile along axial nozzle direction with TBC.



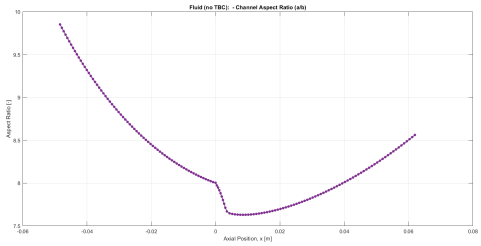
(b) heat transfer coefficient profile along axial nozzle direction with no TBC.

Figure 5.2: Convective heat coefficient profile comparison.

The fluid-side convective heat transfer coefficient is evaluated using the Dittus-Boelter correlation. For a fixed mass flow rate, enhancing convective heat transfer fundamentally requires a reduction in the hydraulic diameter of the cooling channels. Assuming rectangular cross-sections and a fixed channel count ($N_{channels} = 180$ for both configurations), the local channel width (a) is geometrically constrained to the local nozzle circumference divided by the total number of channels (resulting in base widths on the order of millimeters). For rectangular channels, the hydraulic diameter is defined as $D_h = \frac{2ab}{a+b}$, where b represents the local channel height. Since the uncoated configuration necessitates substantially higher convective coefficients, the required channel heights become severely compressed, resulting in high aspect ratio geometries. Figure 5.3 compares the resulting channel aspect ratios (a/b), which should ideally remain low to ensure acceptable manufacturing and structural design limits, highlighting the high aspect ratio required if no thermal barriers are present. The designed channel height, if no TBC are used, falls below 1 mm. The maximum values of the aspect ratio reached are respectively equal to 9.85 for the uncoated and 3.2 for the coated design.



(a) Aspect ratio profile along axial nozzle direction with TBC.



(b) Aspect ratio profile along axial nozzle direction with no TBC.

Figure 5.3: Aspect ratio profile comparison.

5.2 Design of the supporting external apparatus

The external supporting apparatus for the cooling jacket was design for two main reasons. Firstly, it ensures a constant mass flow rate into the system. Secondly, it provides sufficient pressure to prevent vaporization of the water before disposal. The system is design operate in a blowdown configuration, utilizing nitrogen as the pressurant. From the tank to the discharge point, the system comprised the following components in series: a storage tank, a filter, a flow control valve, a flowmeter, a ring-shaped distributor plate, the cooling jacket and a ring-shaped collector plate. Moreover, the system included a thermal well hydraulically modeled as two 90 deg bends to ensure proper temperature measurements. Finally, in the system were considered L=5m with D=5cm of equivalent straight pipes. In the Figure 5.2 is represented a schematic view of how the supporting apparatus was designed.

5.2.1 Computation of pressure losses

Here is proposed a brief description on the calculations of the pressure losses only for the channels in the cooling jacket, as they are the primary source of losses. Finally, a table summarizing all the losses in the system is presented. The complete calculations are in appendix 8.11.

Component	$\boldsymbol{\Delta P}$ [Pa]
Tank	$1.4439 \cdot 10^3$
Filter	$1 \cdot 10^5$
Control Valve	$\approx 0.1 \cdot 10^5$
Flowmeter	≈ 0
Cooling Jacket	$2.4997 \cdot 10^5$
Pipes	$1.5821 \cdot 10^3$
90 deg bends	363.8628
Sharp bend (Mitre)	$8.8426 \cdot 10^3$
Distributor/Collector	209.5079
Margin	20%
Total	$4.4690 \cdot 10^5$

Table 5.1: Hydraulic components and relative pressure losses

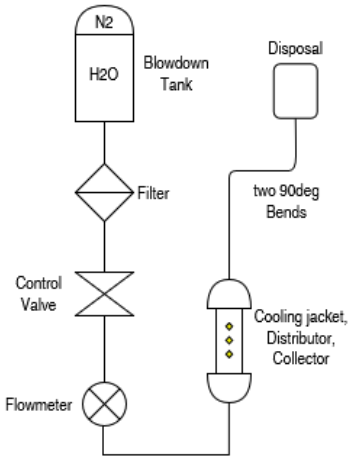


Table 5.2: Simplified schematics of the hydraulic system

Cooling Jacket Although the real cooling jacket presents a complex curvilinear geometry, for the purpose of a hydraulic approximation it was designed as $N = 180$ parallel channels. Each of these channels are approximated as two rough, straight cylindrical tubes with axially varying diameters that correspond to their local hydraulic diameters. These pipes are arranged parallel to the nozzle and joined in the throat, thus forming a $\theta = 68.7716\,\text{deg}$ sharp bend. Therefore, the losses in the cooling jacket are $\Delta p = \Delta p_{irr} + \Delta p_{rev} + \Delta p_{bend}$. The pressure losses in the straight tubes, due to the heat transfer must be computed through integration. Considering the general expression for the irreversible pressure gradient (derivation in appendix 8.10) $-\frac{dp}{dx} = f_D \frac{\rho v^2}{2D}$, where x is the arc length coordinate along the nozzle profile, the evaluation of pressure losses requires accounting for variations in the thermophysical properties of water due to heat transfer along the channels. The density exhibits only minor changes and can therefore be considered constant, while the viscosity may vary significantly. Knowing the local temperature, then the viscosity[16], the Reynold number and the Darcy factor can be computed. To account for viscosity variations in the radial direction of the flow, a corrected friction factor is introduced [32]: $f_{dia} = f(Re(T)) \left(\frac{\mu_b}{\mu_w}\right)^{0.27}$, where μ_b is the bulk viscosity and μ_w the wall viscosity. Furthermore, from the continuity equation, the variation in the diameter of the tubes implies a variation in the velocity, thus a reversible pressure difference between adjacent sections must be computed. The relations are:

$$\Delta P_{irr} = \frac{1}{2}\rho \int_0^L \frac{f_{dia} v(x)^2}{D(x)} dx \qquad \Delta P_{rev} = \frac{\dot{m}^2}{2\rho}\left(\frac{1}{A_{out}^2} - \frac{1}{A_{in}^2}\right) \tag{5.3}$$

where the subscripts "in" and "out" identify the sections parameter at the beginning and the end of the channel, respectively. Finally, it is assumed that the channels have a surface rugosity of $\epsilon = 0.2$ mm, to reflect the typical surface characteristic of additive manufacturing processes, which are commonly employed in the production of these type of jackets.

5.2.2 Water tank sizing

As previously stated the hydraulic system operates in a blowdown configuration. Consequently, the system must be able to ensure an end-of-life water pressure of at least $P_{tank,min} = P_{max} + \Delta P_{tot} = 24.46\text{bar}$. Since the pressure changes during operation the sizing was performed on $P_{tank} = P_{tank,min} B_D - P_{atm} = 72.40\text{bar}$, where P_{tank} is the maximum pressure that the tank must withstand, B_D is the blowdown ratio in this case assumed equal to 3 and P_{atm} the atmospheric pressure. The tank was modeled as a cylindrical shell with hemispherical ends, and it was assumed a ratio $\frac{L}{D} = 3$ for the cylinder. The thicknesses of the tank are computed as [25]:

$$t_{cyl} = \frac{P_{tank,min} R}{S - 0.6 P_{tank,min}} = 10.0130mm \qquad t_{end} = \frac{P_{tank,min} R}{2S - 0.2 P_{tank,min}} = 4.8725mm \tag{5.4}$$

where S is the allowable stress of the material, defined as the yield strength divided by a safety factor of 1.25. Moreover, the mass of the water that the tank is containing was increased by a 20% to account for possible leaks or residual in the lines. The material chosen for the tank was stainless steel 316L [26] characterized by $\rho = 8000\,\text{kg}\,\text{m}^{-3}$ and $\sigma_y = 170\,\text{MPa}$. Once the thicknesses are known the mass of the tank is computed as $M = 119.5268\text{kg}$, while the corresponding tank volume is $V = 0.1390\text{m}^3$ (Calculations in appendix 8.12).

5.2.3 Pressure vessel sizing

A preliminary pressure vessel design of the motor was performed to verify that the selected materials and geometry can withstand the loads produced by the combustion gases, and to provide an initial estimate of the mass. The assumption of the model are reported below. The structure of the motor was approximated as a cylindrical shell with a hemispherical head. The nozzle was assumed equivalent to two truncated cones meeting in the nozzle throat. The convergent was sized considering an internal constant pressure equal to the chamber pressure, while the divergent considering a constant throat pressure. The system was assumed to have ideal joints (weld efficiency $E = 1$), and its material properties were considered unaffected by thermal loads during operation. Finally, all the components were assumed to share a common external diameter, with wall thickness defined inward from the outer surface. Any thickness discontinuities were assumed to be smoothly blended. In the table 5.3 are reported the minimum thicknesses of the various components of the motor. It is to be noted that, the assumption of constant section of the convergent produce an overestimation of the material structurally needed. Consequently, for the purposes of the thermal analysis, in the Table 5.3 was also reported the minimum thickness that the convergent must have at an area ratio $\frac{A}{At} = 2$. (Calculation in Appendix 8.13)

Component	**t**[mm]	**Volume**[m^3]	**Mass**[kg]	**Material**
Shell	13.2071	0.0620	167.5318	Al 6061-T6
Head	6.6036	0.0088	23.8455	Al 6061-T6
Convergent (initial section)	24.6997	0.0214	184.1358	GRCop-84
Convergent ($\frac{A}{At} = 2$)	3.8441	/	/	GRCop-84
Divergent	3.2382	0.0014	12.0713	GRCop-84

Table 5.3: Thickness, Volume, Mass and the Material considered for each component

Although the minimum thickness varies along the nozzle axis, for the purposes of the thermal analysis it was assumed constant, with a value of $t_{thermal} = 10mm$, which exceeds the minimum required thickness. The material selected for the case was alluminum 6061-T6 [37], which has a good density to yield ratio while remaining relatively cheap. For the nozzle components GRCop-84 [29]was chosen[1], thanks to its thermal proprieties. A safety factor of 1.15[36] was imposed to the yield stresses.

5.2.4 Sensors selection for flow temperature measurement

The Automated Logic N1-10K-2 immersion thermistor series was selected for coolant temperature acquisition at the inlet and outlet water collectors of the rocket engine cooling circuit. [39] The primary measurement objective is to determine the coolant temperature difference across the engine cooling jacket water input and output sections, from which the heat flux absorbed by the cooling jacket during a motor test-fire can be derived via calorimetric calculation. Given the expected coolant operating range of approximately 18–70°C (depending on water mass flow rate), the NTC thermistor element is preferred over thermocouple-based sensing. The thermistor provides an accuracy of ±0.2°C (±0.1°C with the [XP] option), yielding a worst-case uncertainty of ±0.4°C when differencing two matched sensors. Probe length (2", 4", or 8") can be selected to match the wanted thermo-well insertion depth, ensuring adequate immersion for representative bulk fluid temperature measurem ent.

[1]The yield strength was tanken as 200MPa, corresponding to $T \approx 500K$

Chapter 6: Model Limitations and Possible Improvements

Frozen mixture model All NASA CEA simulations use a frozen mixture approach. While necessary to ensure internal consistency, this leads to an underestimation of both motor performance and nozzle flow temperatures. Adopting an equilibrium composition model with a validated set of transport properties would improve fidelity, provided internal consistency is maintained.

Unmodeled nozzle losses Divergence losses were accounted for, however, viscous and two-phase flow losses were not. Viscous losses, primarily the boundary-layer-induced reduction of effective throat area, are typically difficult to quantify at the preliminary design stage. Two-phase flow losses, can reach approximately 5% of I_{sp} in aluminised formulations but are expected to be negligible for the non-aluminised propellant used here.

Uniform burning rate assumption The burning rate r_b is assumed spatially uniform over all active grain surfaces at any given instant, neglecting axial pressure gradients and local regression-rate variations due to gas velocity or thermal effects. This simplification is consistent with standard BATES data reduction practice and is justified by the high port-to-throat area ratio ($A_p/A_t \approx 20$), which suppresses both erosive burning and meaningful axial pressure drops. A finer grain mesh discretisation to account for local regression rate differences would improve fidelity, though this level of depth goes beyond the purposes of the presented preliminary analysis.

Ignition modelling The ignition transient is represented by imposing an initial chamber pressure of 10 bar, assuming instant flame propagation over the grain surface. An accurate modelling of the igniter's effect and of the flame diffusion process would improve simulation fidelity.

Limited uncertainty propagation The Monte Carlo campaign propagates uncertainty only on the Vieille law parameters a and n. A more comprehensive study could include variability in propellant density, geometrical parameters, throat erosion rate, combustion efficiency η_{c^*}, and nozzle divergence losses.

Single-segment grain geometry The current single-segment design satisfies the neutrality condition at ignition and burnout but exhibits a mildly progressive-regressive behaviour. A multi-segment BATES configuration would offer additional degrees of freedom to flatten the pressure trace.

Steady-state thermal assumption The heat transfer analysis assumes steady-state conditions throughout the burn. For a 25 s firing, the thermal wave may not fully penetrate the nozzle wall and TBC layers, meaning the actual wall temperatures could be lower than the computed steady-state values. The present approach is therefore conservative:

it overestimates both the wall temperature and the heat flux that the cooling jacket must absorb. A transient heat transfer analysis could potentially relax TBC thickness requirements.

TBC survivability justification The current TBC survivability argument relies on a qualitative comparison with published cycling data; a dedicated thermomechanical stress analysis would place this conclusion on firmer ground.

Planar wall approximation Each panel of the thermal model is treated as a planar composite wall, neglecting the curvature of the nozzle contour. In the throat region, where the radius of curvature is smallest, the cylindrical geometry introduces a difference between inner and outer surface areas that is not captured by the planar assumption.

Constant thermophysical properties across layers Thermal conductivities of the structural wall, bond coat, and ceramic layers are assumed constant through their respective thicknesses. In reality, the steep temperature gradient across the layers implies significant property variations.

Neglected axial conduction The panel method resolves heat transfer only in the wall-normal direction. Axial conduction along the nozzle wall, which would tend to smooth the temperature profile in the throat region by redistributing heat toward cooler upstream and downstream sections, is not modelled. This omission is conservative for the peak throat temperature but means the predicted axial temperature gradients may be sharper than in reality.

Uniform coolant distribution The cooling jacket model assumes equal mass flow rate across all channels. In practice, the ring-shaped distributor and collector will introduce channel-to-channel flow imbalance, potentially creating localised hot spots that are not predicted by the present analysis. A more detailed hydraulic network model, or a CFD simulation of the distributor, would help to partially quantify this effect.

Neglected thermal stress contributions Structural sizing of the nozzle and pressure vessel was performed considering only pressure loads. Thermal stresses arising from the steep radial temperature gradient across the wall assembly, particularly at the TBC-substrate interface, were not evaluated. Adequate margins were imposed on the pressure-based sizing to partially compensate for this omission, but a dedicated thermomechanical analysis would be required to confirm structural integrity.

Neglected covariance in Monte Carlo dataset generation The Monte Carlo campaign samples a and n as independent Gaussian variables. However, since both parameters are derived from the same linear regression, their estimators are statistically correlated. Neglecting this correlation and sampling from independent marginals effectively overpopulates regions of the (a, n) space that the regression would assign low joint probability, broadening the predicted performance distributions.

Chapter 7: Conclusions

Summary of work performed The present work covered the full preliminary design cycle of the Stark-100k: propellant characterisation via Vieille law regression from 15 small-scale BATES firings; grain geometry sizing through simultaneous imposition of neutrality, mass-balance, and total-impulse constraints; performance verification via both quasi-steady and unsteady (RK4 + NASA CEA) chamber pressure solvers; a Monte Carlo uncertainty campaign on the Vieille parameters a and n; Rao bell nozzle profiling with equivalent divergence losses; ablative thermal protection sizing for the combustion chamber and nozzle extremities; selection and justification of a multilayer TBC (RSZ/YSZ/NiCrAlY) for the throat region; a steady-state conjugate heat transfer analysis via Bartz convection and a multi-layer panel method, including a quantitative comparison with and without TBC; cooling jacket channel sizing with Dittus–Boelter correlation; hydraulic loss computation across the full blowdown circuit; water tank and pressure vessel preliminary structural sizing per ASME BPVC; and sensor selection for calorimetric heat flux measurement.

Design process output The designed Stark-100k motor is compliant with the design performance requirements. The designed cooling jacket is also compliant with the evaluated material survivability requirements, if coupled with the designed TBC. All dimensioned components fall within acceptable ranges and are coherent with existing technology solutions. Preliminary design for external apparatus was successful.

Lessons learned The cooling jacket design proved to be the most challenging aspect of the project, requiring a careful balance between material thermomechanical properties, heat flux requirements, and coolant-side constraints within a highly coupled design space. Specifically, the strong nonlinearity of the governing relations meant that some controlled variables were nearly insensitive to certain design degrees of freedom while being disproportionately sensitive to others, making manual iteration through the design space difficult.
Avoiding degenerate cooling channel aspect ratios (e.g. highly stretched rectangular ducts) was particularly important to maintain adequate manufacturability and avoid extreme pressure losses due to excessive "tightness" of the channels.
Another notable observation is that pressure losses across the cooling circuit are highly non-linear in the imposed mass flow rate $\dot{m}$. This behavior is a reminder that in a highly coupled design space, intuitive scaling relationships often do not hold.

On the ballistic side, the close agreement between the quasi-steady and unsteady solvers (often $\sim$1% on performance metrics) was closer than initially expected, given the added complexity of the unsteady model. This suggests that, for the designed near-neutral BATES grain with a high port-to-throat area ratio, the chamber filling dynamics have a negligible effect on integrated performance and the additional computational cost of an unsteady simulation is justified primarily for resolving peak-pressure compliance near the MEOP ceiling rather than for improving average performance predictions.
Finally, the design process highlighted the big impact of TBCs on the system's viability, as demonstrated in the comparison between coated and uncoated configurations.

Next steps and important considerations for future work The natural progression for the expansion of the presented work involves the transition from the current baseline architecture towards an higher-fidelity modeling. While all considerations expressed in the previous chapters are deemed correct, two areas warrant priority attention. First, the structural sizing of both the Stark-100k motor and the cooling jacket accounts only for pressure loads, neglecting the thermal stresses generated by the steep radial temperature gradient across the multilayer wall assembly. A coupled thermomechanical analysis would either confirm the current margins or identify the need for redesign.
Secondly, the thermal model captures only convective heat transfer via the Bartz correlation, omitting radiative emission from combustion products in the throat region. This unmodeled contribution would increase the total wall heat flux, potentially affecting both the TBC temperature margin and the cooling jacket sizing. Moreover, in order to accurately size ablative materials, a transient high-fidelity model, to couple nozzle heat flux and surface regression due to ablation should be employed for performance assessment.

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Chapter 8: Appendices

8.1 Rao nozzle

The Rao nozzle is designed starting from the corresponding conical nozzle with angle of the divergence of $\alpha = 15°$, the fraction of the length of the Rao nozzle with respect to the equivalent conical nozzle of reference is computed in order to have similar divergence losses in the two nozzles, therefore coefficient λ has to be the same. The initial data imposed to define the geometry are: the angle of the convergent part (β), the angle of the divergent part (α), the throat and exit areas (A_t, A_e) and the radius of the combustion chamber (R_{cc}). From these quantities some other dimensions are retrieved with some geometric analytical relations:

- throat diameter: $d_t = 2\sqrt{\frac{A_t}{\pi}}$
- exit diameter: $d_e = 2\sqrt{\frac{A_e}{\pi}}$
- throat radius: $R_t = \frac{d_t}{2}$
- exit radius: $R_e = \frac{d_e}{2}$
- length of the convergent part of the conic nozzle $L_{c,\text{conic}} = \frac{R_{cc} - \frac{d_t}{2}}{\tan(\beta)}$
- length of the divergent part of the conic nozzle $L_{d,\text{conic}} = \frac{R_e - R_t}{\tan(\alpha)}$
- length of the divergent part of the conic nozzle with $\alpha = 15°$: $L_{d,\text{conic15}} = \frac{R_e - R_t}{\tan\left(\frac{\pi}{12}\right)}$
- $L_{\text{nozzle}} = L_{c,\text{conic}} + L_{d,\text{conic}}$

Then, the divergence losses coefficient for the equivalent conical nozzle can be calculated as shown in 8.1.

$$\lambda = \frac{1 + \cos(\alpha)}{2} \tag{8.1}$$

For different values of the percentage of the length of the Rao nozzle with respect to the equivalent conical nozzle (η), the losses of the Rao nozzle (eq 8.3) are compared with the result of 8.1, in which α_1 is retrieved from 8.2.

$$\alpha_1 = \tan^{-1}\left(\frac{R_e - R_t}{\eta L_{d_{\text{conic15}}}}\right) \tag{8.2}$$

$$\lambda_{\text{RAO}} = \frac{1}{2}\left(1 + \cos\left(\frac{\alpha_1 + \theta_e}{2}\right)\right) \tag{8.3}$$

The values of θ_i and θ_e can be extracted from the graph shown in 8.1. However, since the graph does not include curves for every value of the percentage of the Rao nozzle length with respect to the conical nozzle at $\alpha = 15$ a function as been implemented to find the values of the chracteristic angles of the divergent. In this function the missing values

of the graph has been interpolated basing on the values that are already in the graph. Moreover, in 8.2 it is shown the actual graph used by the implemented function. By the colorbar, it is possible to distinguish the different contour lines corresponding to different percentage values of the conical divergent length with a 15° alpha angle. The upper lines correspond to θ_i, while the lower lines correspond to θ_e.

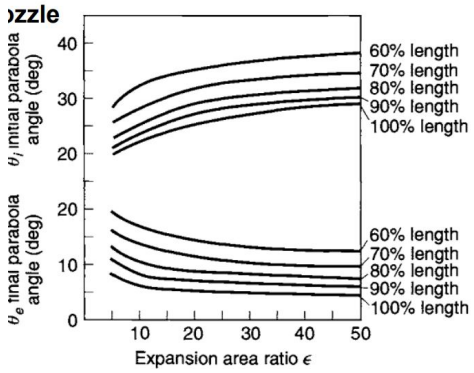


Figure 8.1: Rao angles

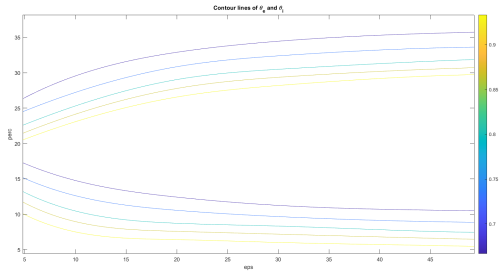


Figure 8.2: plot estimated Rao angles

The geometry of the nozzle is fixed by Rao model which contour is approximated using a classical bell approach. The different curves are matched from the exit of the combustion chamber to the exit of the nozzle: a line two circumferences, a parabola with axis parallel to the direction of the length of the nozzle. The geometry is continuous in slope at patching points.

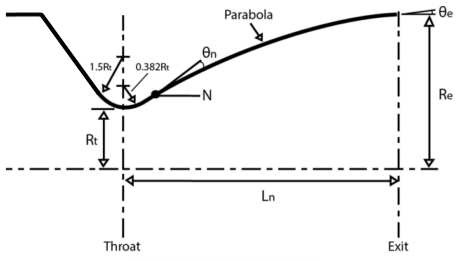


Figure 8.3: Geometric constraints of Rao nozzle

$$r_{RAO}(x) = \begin{cases} mx + q & x_0 \leq x < x_1 \\ 1.5R_t + \sqrt{(1.5R_t)^2 - x^2} & x_1 \leq x < x_2 \\ 0.382R_t - \sqrt{(0.382R_t)^2 - x^2} & x_2 \leq x < x_3 \\ \frac{-b+\sqrt{b^2-4a(c-x)}}{2a} & x_3 \leq x \leq x_4 \end{cases}$$

The coefficients m, q, a, b, c are computed by imposing Rao geometrical contraints .The resulting C^1 profile can be therefore differentiated with respect to x, and the expression of the tangent and normal unit vectors at each x on the curve can be computed as

$$\hat{t}(x) = \frac{1}{\sqrt{1 + (r'_{RAO}(x))^2}} \begin{pmatrix} 1 \\ r'_{RAO}(x) \end{pmatrix}, \hat{n}(x) = \frac{1}{\sqrt{1 + (r'_{RAO}(x))^2}} \begin{pmatrix} -r'_{RAO}(x) \\ 1 \end{pmatrix} \tag{8.4}$$

Also, it can be integrated to obtain the curvilinear length of the shape at each x:

$$l_{RAO}(x) = \sum_{i=1}^{n=4} \int_{x_{i-1}}^{x_i} \sqrt{1 + \left(\frac{dr_i(x)}{dx}\right)^2} dx \quad \in [0, L]$$

Curvilinear discretization To ease the computational effort, a finite element approach has been implemented, rather than a fully analytical one. Therefore a discretization of the profile into N-1 curvilinear arcs of same curvilinear length dl has been implemented. The nodes are therefore located in curvilinear abscissas

$$l_{RAO,i} = \frac{L}{N}, \quad i = 0, 1, \ldots, N \tag{8.5}$$

where L_{tot} is the total curvilinear length, of the nozzle profile:

$$L = \sum_{i=1}^{n} \int_{x_{N_{in}}}^{x_{N_{out}}} \sqrt{1 + \left(\frac{dr_i(x)}{dx}\right)^2} dx$$

where $x_{N_{in}} = x_0$ and $x_{N_{out}} = x_4$ are the axial abscissas at the threshold between combustion chamber and nozzle convergent and at exhaust. It is then possible to compute the axial abscissas of each node as $x_i = x(l_{RAO})$, thanks to the previously remarked invertibility of $l(x)$: $\underline{\mathbf{x}}_{nodes} = [x(l_{RAO,_1} = 0) = x_0, x(l_{RAO,_2}), ..., x(l_{RAO,_N} = L) = x_4)]$. And, therefore also the r coordinates of the nodes are obtained, as $\underline{\mathbf{r}}_{nodes} = r_{RAO}(\mathbf{x}_{nodes})$.

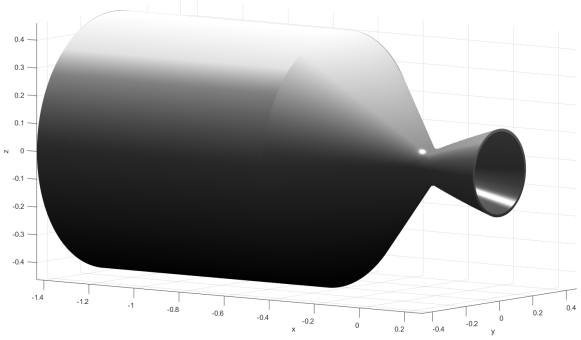


Figure 8.4: Rao nozzle 3D render.

8.2 Thermal barrier safety thickness

The thickness chosen for the ablative material

$$t_{Ablative} = S_f \left(\dot{r} \Delta t_{burn} + \sqrt{\alpha_\perp \Delta t_{burn}} \right) \tag{8.6}$$

considers an additional safety thickness of

$$\delta = \sqrt{\alpha_\perp \Delta t_{burn}} \quad . \tag{8.7}$$

The reasoning behind this choice is rooted in the solution of the differential problem of the heat exchange inside a semi-infinite wall due to a step temperature offset. The non-steady heat conduction equation for this problem is Fourier equation, that in this case becomes:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad . \tag{8.8}$$

While in our case $\alpha = \alpha_\perp$, considering the direction in which the temperature is propagated inside the ablative material. This is the well-known diffusivity equation also describing other physical effects in similarity to this one in other continuous physical domain problems, such as the First and Second Stokes Theorem in Fluid Dynamics. By introducing the similarity variable $\eta = \frac{x}{\sqrt{4\alpha t}}$, we apply the chain rule:

$$\frac{\partial T}{\partial t} = \frac{dT}{d\eta}\frac{\partial \eta}{\partial t} = -\frac{1}{2}\frac{x}{\sqrt{4\alpha t^3}}\frac{dT}{d\eta} = -\frac{\eta}{2t}\frac{dT}{d\eta} \tag{8.9}$$

$$\frac{\partial T}{\partial x} = \frac{dT}{d\eta}\frac{\partial \eta}{\partial x} = \frac{1}{\sqrt{4\alpha t}}\frac{dT}{d\eta} \implies \frac{\partial^2 T}{\partial x^2} = \frac{1}{4\alpha t}\frac{d^2 T}{d\eta^2} \tag{8.10}$$

Substituting these into the original PDE yields the ODE:

$$-\frac{\eta}{2t}\frac{dT}{d\eta} = \alpha \left(\frac{1}{4\alpha t}\frac{d^2 T}{d\eta^2} \right) \implies \frac{d^2 T}{d\eta^2} + 2\eta \frac{dT}{d\eta} = 0 \tag{8.11}$$

Solving by successive integration with boundary conditions $T(0) = T_w$ and $T(\infty) = T_0$:

$$\frac{T(x,t) - T_0}{T_w - T_0} = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-u^2} du = 1 - \mathrm{erf}(\eta) = \mathrm{erfc}\left(\frac{x}{\sqrt{4\alpha t}} \right) \tag{8.12}$$

Therefore, the justification of $\delta(\alpha, \Delta t_{burn})$ as safety thickness follows quite immediately. Indeed, the function $\mathrm{erfc}(\eta)$ decays exponentially. Choosing a thickness proportional to $\delta \approx \sqrt{\alpha \Delta t_{burn}}$ ensures that the thermal perturbation has in good approximation not yet reached the cold structure at $x > \delta$, as the temperature gradient is virtually extinguished at that distance.

8.3 Erosive burning model

$$r = r_0 + r_e = a p_1^n + \alpha G^{0.8} D^{-0.2} \exp(-\beta r \rho_b / G) \tag{8.13}$$

$$\alpha = \frac{0.0288 c_p \mu^{0.2} \mathrm{Pr}^{-2/3}}{\rho_b c_s} \frac{T_1 - T_s}{T_2 - T_b} \tag{8.14}$$

Where:

- β is an empirical constant, usually around 53 [22]
- r_0 is the burning rate in the absence of gas flow (quantified via Vieille's Law: ap^n).
- G the mass flow velocity per unit area.
- ρ_b is the density of the unburned propellant.
- D is a characteristic dimension of the port passage (usually, D = $4A_p/S$, where Ap is the port area and S is its perimeter).
- c_p and c_s are the average specific heat at constant pressure of the combustion gases and solid propellant respectively. [23]
- μ is the gas viscosity and Pr is the Prandtl number.
- T_1 is the flame temperature, $T_s \approx 800$ K is the propellant burning surface temperature [23], T_2 is the gas core temperature in the port, and T_b is the bulk (initial) propellant temperature. [24]

8.4 Pressure simulation: combustion chamber dynamics model

The objective is to retrieve a differential equation that can render the combustion chamber dynamics modeling possible in a general way, without on relying predetermined mean pressure values. Given a generic control volume V it is possible to impose a mass conservation equation:

$$\frac{d(\rho V)}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

where $\dot{m}_{in}$ e $\dot{m}_{out}$ represent the mass flow rates that enter and exit the control volume respectively.

Under the hypothesis of perfect gasses in the combustion chamber, where P is the pressure and R is the universal gas constant, M_w is the molar mass, T is the combustion chamber gas temperature and V is the internal volume of the combustion chamber:

$$\frac{d(\rho V)}{dt} = \rho\frac{dV}{dt} + V\frac{d\rho}{dt}$$

$$\rho = \frac{P_{ch}}{\frac{R}{M_w}T}$$

With A_b indicating the combustion area and r_b indicating the burning rate:

$$\rho\frac{dV}{dt} = \rho A_b r_b$$

$$V\frac{d\rho}{dt} = V\frac{1}{\frac{R}{M_w}T}\frac{dP_{ch}}{dt} - \frac{P_{ch}}{\frac{R}{M_w}T^2}\frac{dT}{dt} + \frac{P_{ch}}{RT}\frac{dM_w}{dt}$$

With $\frac{dT}{dt} << 1$, $\frac{1}{T^2} << 1$, $\frac{dM_w}{dt} << 1$

$$V\frac{d\rho}{dt} = V\frac{1}{\frac{R}{M_w}T}\frac{dP_{ch}}{dt}$$

We can subsequently derive the quasi-steady combustion chamber formulation:

$$V\frac{1}{\frac{R}{M_w}T}\frac{dP_{ch}}{dt} + \rho A_b r_b = \dot{m}_{in} - \dot{m}_{out}$$

Considering that $\dot{m}_{in} = \rho_p A_b r_b$ where ρ_p is the grain density and $\dot{m}_{out} = \frac{P_{ch} A_{gola}}{c^*}$

$$c^* = \sqrt{\frac{RT_{ch}}{\gamma M_w}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

It is therefore possible to write a differential relation between combustion chamber pressure, gemetric parameters and thermodynamic parameters of the system:

$$\frac{V}{\frac{R}{M_w}T}\frac{dP_{ch}}{dt} = [(\rho_p - \rho)A_b r_b - \frac{P_{ch}A_t}{c^*}]$$

$$\frac{dP_{ch}}{dt} = (\frac{V}{\frac{R}{M_w}T})^{-1}[(\rho_p - \rho)A_b r_b - \frac{P_{ch}A_t}{c^*}]$$

That can be expressed as:

$$\frac{dP_{ch}}{dt} = f(P_{ch}, t)$$

With $rb = rb(P_{ch}, T_{ch})$, $A_b = A_b(r_b, t)$, $V = V(r_b, t)$.

Using NASA CEA it is possible to retrieve ρ, T_c, c^*, M_w for a given P_{ch} combustion chamber pressure, considering A_t a known parameter:

$$\rho = \rho(P_{ch})_{CEA}$$
$$c^* = c^*(P_{ch})_{CEA}$$
$$M_w = M_w(P_{ch})_{CEA}$$

8.5 Numerical integration

8.5.1 Nominal design - numerical integration strategy

Given the relation

$$\frac{dP_{ch}}{dt} = (\frac{V}{\frac{R}{M_w}T})^{-1}[(\rho_p - \rho)A_b r_b - \frac{P_{ch}A_t}{c^*}]$$

That can be expressed as:

$$\frac{dP_{ch}}{dt} = f(P_{ch}, t)$$

It is possible to preform its numerical integration using a classical fourth order Runge-Kutta method.
During the nominal design P_{ch} dynamics simulation, each thermodynamic property is evaluated by solving a CEA rocket problem for each intermediate RK4 integrator step, meaning 4 CEA rocket problems were solved for each integration step.
Each geometrical property was evaluated using a discrete grain section regression model. Specifically, a grain section [1] mesh is generated upon simulation start, composed of vectors containing the coordinates of its vertices. The regression of the generated mesh is then modeled by a simulation coupled with the combustion chamber dynamics integrator, and is used to compute both the A_b and $V_{chamber}$ values at each integrator substep.

8.5.2 Montecarlo simulation campaign - numerical integration strategy

A slight variation of the numerical integration strategy described in 8.5.1 was employed during the Montecarlo campaign simulations: in order to significantly reduce runtime, each CEA-retrieved combustion chamber gas thermodynamic property was assumed to be constant during a single RK4 timestep. Therfore, the four RK4 substeps performed during each integration step all used the results of a single CEA rocket problem, significantly reducing computational overhead. This integration method was validated against the complete RK4 integrator described in 8.5.1, and showed meaningless differences in terms of simulation results.

8.5.3 Initial conditions

The initial condition used to start the numerical integration process was assumed to be $P_{start} = 10$ bar, intended to be representative of the effect of an igniter. This specific pressure value vas chosen to keep the dynamical process within its theoretical validity limits. Specifically, this specific initial pressure value allows the $\frac{dT}{dt} << 1$ assumption to remain satisfied.
The employment of this initial condition implies the assumption of instant flame propagation over the grain surface.

[1] to be intended as a grain section lying on the r, z plane, where z is the motor's longitudinal axis and r a given radial direction

8.5.4 Chamber deflation modeling

The differential equation described in 8.4 was used to model chamber deflation process. During this second phase of integration the value of $\frac{dT}{dt}$ gets significantly close to 1, undermining the accuracy of the obtained numerical results.
For this reason the deflation process contribution to the I_{tot} value was chosen to be ignored.

8.6 Optimal length in BATES motors

The general burning surface is given by:

$$A_b = A_{b_{cylinder}} + 2A_{b_{end}} = \pi dl + \frac{\pi}{2}\left(D^2 - d^2\right) \tag{8.15}$$

where D is the outer diameter while d and l the inner diameter and total length at a given time respectively.
The instantaneous quantities can be expressed as a function of the burned thickness w, which is the same in the radial and axial direction.

$$d = d_0 + 2w \quad \text{and} \quad l = L - 2w$$

where L is the initial grain length and d_0 the initial port diameter.
Substituting these relations in eq (8.15) and simplifying yields:

$$A_b(w) = \pi\left(d_0 L - \frac{d_0^2}{2} + \frac{D^2}{2} + 2wL - 4wd_0 - 6w^2\right) \tag{8.16}$$

To impose neutrality the initial area and the final area are considered the same:

$$\begin{aligned} A_b(0) &= \pi\left(d_0 L - \frac{d_0^2}{2} + \frac{D^2}{2}\right) = A_b(w_f) = \\ &= \pi\left(d_0 L - \frac{d_0^2}{2} + \frac{D^2}{2} + (D-d_0)L - 2d_0(D-d_0) - 3(D-d_0)^2\right) \end{aligned}$$

Where $w_f = \frac{(D-d_0)}{2}$ was used.

Simplifying the terms gives the condition for the neutrality of the grain:

$$L = \frac{3}{2}D + \frac{d_0}{2} \tag{8.17}$$

8.7 c* for BATES motors

Starting from the following relation

$$\dot{m}_{in} = \dot{m}_{out} + \frac{d}{dt}\left(\frac{P_c M_m}{RT_c} V_c\right) \tag{8.18}$$

and considering a constant pressure and, consequently, temperature and small variations of V_c with respect of the other quantities.
The derivative can be neglected, obtaining:

$$\dot{m}_{in} = \dot{m}_{out} = \frac{P_c A_t}{c^*} \quad \text{and so} \quad c^* = \frac{P_c A_t}{\dot{m}_{in}}$$

However, measuring the flow rate of a rocket motor is a rather complicated task; to solve for this, the flow rate is integrated over the burning time

$$m_p = \int_0^{t_b} \frac{P_c A_t}{c^*} dt \tag{8.19}$$

Being the throat area and the characteristic velocity constants (no or negligible throat erosion), they can be removed from the integral. Inverting then the relation yields an expression for c^*:

$$c^* = \frac{A_t}{m_p} \int_0^{t_b} P_c \, dt \tag{8.20}$$

where m_p is the burned propellant mass (if residue are present they must be accounted for) and t_b the total burn time.
With pressure transducers and thanks to the grain neutrality, P_c can be easily read, numerically integrated and so c^* experimentally recovered. Comparing it to the theoretical value form thermochemistry, the efficiency η_{c^*} is computed.

8.8 Heat transfer thermodynamic problems

The convective heat transfer coefficient, h_{HotGas}, is computed according to the Bartz model, as $h_{HotGas}(T_{WallGas})$

$$h_{HotGas} = \frac{0.026}{D_t^{0.2}} \left(\frac{D_t}{R_{c,t}}\right)^{0.1} \frac{\mu_0^{0.2} c_p}{Pr^{0.6}} \left(\frac{p_0}{c^*}\right)^{0.8} \left(\frac{A_t}{A}\right)^{0.9} \sigma(M, T_{wallGas}, T_0) \tag{8.21}$$

where σ represents the property variation correction factor. This term accounts for the steep temperature and density gradients across the boundary layer and is defined as [19]:

$$\sigma = \left[\frac{1}{2}\frac{T_{wallGas}}{T_0}\left(1 + \frac{\gamma - 1}{2} M^2\right) + \frac{1}{2}\right]^{-0.68} \left[1 + \frac{\gamma - 1}{2} M^2\right]^{-0.12} \tag{8.22}$$

8.9 Water properties

Dynamic Viscosity of Water

The dynamic viscosity $\mu(T)$ is modeled using an empirical exponential correlation. This model is specifically calibrated for liquid water within a temperature range of approximately 273.15 K to 373.15 K at atmospheric pressure. The output unit is Pascal-seconds ($\text{Pa} \cdot \text{s}$), equivalent to $\text{kg}/(\text{m} \cdot \text{s})$.

$$\mu(T) = 1.856 \cdot 10^{-14} \exp\left(\frac{4209}{T} + 0.04527\,T - 3.376 \cdot 10^{-5}\,T^2\right) \text{[16]} \tag{8.23}$$

The coefficients in the exponent are derived to work with absolute temperature T expressed in Kelvin (K).

Thermal Conductivity

The thermal conductivity $K(T)$ is determined using the formulation by Ramires et al. (1995). It relies on a dimensionless scaling factor $\eta(T)$, which represents the deviation from a reference state. This model is the international standard for the conductivity of liquid water between 274 K and 370 K.

$$\eta(T) = -1.48445 + 4.12292 \left(\frac{T}{298.15} \right) - 1.63866 \left(\frac{T}{298.15} \right)^2 \tag{8.24}$$

$$K(T) = \eta(T) \cdot 0.6065 \text{ [17]} \tag{8.25}$$

The resulting unit is Watts per meter-Kelvin ($\mathrm{W/(m \cdot K)}$).

Specific Heat Capacity

The isobaric specific heat capacity $c_p(T)$ is calculated using the Shomate equation as defined by the NIST Chemistry WebBook. This polynomial fit is highly accurate for liquid water under saturation pressure. The coefficients A, B, C, D, E are specific to water, and the result is divided by the molar mass $M_{m,\text{water}} = 0.018015\,\mathrm{kg/mol}$ to convert the value from molar to mass-specific units ($\mathrm{J/(kg \cdot K)}$).

$$c_p(T) = \frac{A + B\tau - C\tau^2 + D\tau^3 + E\tau^{-2}}{M_{m,\text{water}}} \text{ [18]} \tag{8.26}$$

Where $\tau = T/1000$ (with T in Kelvin) and the constants are:

- $A = -203.6060, B = 1523.290, C = 3196.413, D = 474.455, E = 3.855326$

Dimensionless Transport Numbers

To evaluate the heat transfer efficiency within the calculated volume, the following dimensionless groups are defined:

- **Reynolds Number (Re):** Represents the flow regime (laminar vs. turbulent).

$$Re(T, v, D) = \frac{\rho\, v\, D}{\mu(T)} \tag{8.27}$$

- **Prandtl Number (Pr):** Relates the momentum boundary layer to the thermal boundary layer.

$$Pr(T) = \frac{c_p(T)\, \mu(T)}{K(T)} \tag{8.28}$$

- **Nusselt Number (Nu_{Db}):** Defined by the Dittus-Boelter correlation for turbulent heating in pipes.

$$Nu_{\text{Db}}(Re, Pr) = 0.0243\, Re^{0.8}\, Pr^{0.4} \tag{8.29}$$

8.10 Pressure Gradient in rectilinear tubes

To derive the pressure gradient expression, we apply the momentum balance to an infinitesimal control volume of length dx, cross-sectional area A, and wetted perimeter P.

8.10.1 Force Balance

According to Newton's Second Law for a steady flow, the sum of forces acting on the fluid element equals the change in momentum flux:

$$\sum F = \dot{m} \cdot dv \tag{8.30}$$

The forces acting on the element are:

- **Upstream pressure force:** pA
- **Downstream pressure force:** $-(p + dp)(A + dA)$
- **Wall pressure component:** pdA (projected in the flow direction)
- **Wall shear (friction) force:** $-\tau_w P dx$

Substituting these into the balance equation:

$$pA - (p + dp)(A + dA) + pdA - \tau_w P dx = \rho A v \, dv \tag{8.31}$$

Expanding the product $(p+dp)(A+dA)$ and neglecting the second-order term $dpdA \approx 0$:

$$pA - (pA + pdA + Adp) + pdA - \tau_w P dx = \rho A v \, dv \tag{8.32}$$

Simplyfing:

$$-Adp - \tau_w P dx = \rho A v \, dv \tag{8.33}$$

Dividing the entire equation by A:

$$-dp - \tau_w \frac{P}{A} dx = \rho v \, dv \tag{8.34}$$

Introducing the following definitions:

- **Hydraulic Diameter (D):** $D = \frac{4A}{P} \implies \frac{P}{A} = \frac{4}{D}$
- **Wall Shear Stress (τ_w):** $\tau_w = \frac{f}{8}\rho v^2$ (where f is the Darcy friction factor)

Substituting these into the simplified balance:

$$-dp - \left(\frac{f}{8}\rho v^2\right) \frac{4}{D} dx = \rho v \, dv \tag{8.35}$$

$$-dp - \frac{f}{D}\frac{\rho v^2}{2} dx = \rho v \, dv \tag{8.36}$$

Dividing by dx and isolating the pressure gradient dp/dx:

$$\frac{dp}{dx} = - \underbrace{\frac{f}{D}\frac{\rho v^2}{2}}_{\text{Frictional term}} - \underbrace{\rho v \frac{dv}{dx}}_{\text{Acceleration term}} \tag{8.37}$$

Where:

- The **frictional term** represents the pressure drop due to viscosity.
- The **acceleration term** represents the pressure change due to the varying cross-section (Bernoulli effect).

To find the finite pressure change (Δp_{acc}) between an initial section A_1 and a final section A_2, the acceleration term is integrated:

$$\Delta p_{acc} = \int_{v_1}^{v_2} \rho v \, dv = \frac{\rho}{2}(v_2^2 - v_1^2) \tag{8.38}$$

Substituting $v = \dot{m}/(\rho A)$, ab expression in terms of mass flow rate is obtained:

$$\Delta p_{acc} = \frac{\dot{m}^2}{2\rho}\left(\frac{1}{A_2^2} - \frac{1}{A_1^2}\right) \tag{8.39}$$

8.11 Pressure Losses

Tank, Pipes and Bends

From the tank two types of pressure losses were considered. First, as the stationary water inside the tank accelerates in the pipes, part of the initial static pressure is converted into kinetic energy through a reversible process. Then as the flow pass through the exit orifices of the tank, turbulence along with vena-contracta effect generate irreversible pressure loss. These phenomena are captured in the following equation:

$$\Delta P_{tank} = \frac{1}{2}\rho\left(\frac{v_p}{Cd}\right)^2 \tag{8.40}$$

where v_p^2 is the fluid velocity in the pipes (assumed constant in the section) and Cd is the discharge coefficient. In this case the tank opening was considered as a orifice with $Cd = 0.6$.
The pipes losses (excluding the cooling jacket itself) were computed using the typical relation $\Delta P = \frac{1}{2} f \left(\frac{L}{D}\right) \rho v^2$ where f is the Darcy friction factor, L the pipe length and D its hydraulic diameter. The pipes were considered rectilinear, adiabatic and with constant area and the water properties were also considered constant during the flow. Particularly $\rho = 998 \mathrm{kg\,m^{-3}}$ and $\mu = 1.053 \cdot 10^{-3} \mathrm{Pa\,s}$
The Darcy factor was computed using two different models, in function of the Reynolds number: the Hagen–Poiseuille relation for laminar flow, and the Haaland relation for turbulent flow.

$$f = \frac{64}{Re} \quad Re < 2000 \qquad \frac{1}{\sqrt{f}} = -1.8 \log\left(\left(\frac{\epsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right) \quad Re > 2000 \tag{8.41}$$

Where ϵ is the rugosity of the pipe material.
In the configuration were also modeled two smooth 90 deg bends. These are situated in the piping after the cooling jacket and they are part of a thermal well configuration to allow for precise temperature measurements. Assuming a relative curvature of the bends equal to 2 then the loss coefficient is $K_{smooth} = 0.35$ [30]. The pressure loss coefficient for the Mitre bend was given by the relation [30]:$K_{sharp} = 0.95 \sin^2 \frac{\theta}{2} + 2.05 \sin^4 \frac{\theta}{2}$, and the pressure loss was calculated considering the mean velocity of the flow in the channels.

Mass rate valve, Filter and Flowmeter

The flow control valve, was assumed to have a constant flow coefficient corresponding to a ball valve # 150 with a $\frac{1}{2}$" diameter fully opened [31]. From this data in $Cv = 20$, so $Kv = 17.3$. The pressure losses in the case of water are $\Delta p = \left(\frac{\dot{m}}{K_v}\right)^2 \approx 0.1$bar, where $\dot{m}$ is measured in $\mathrm{m^3\,h^{-1}}$. For the filter a conservative pressure loss value of $\Delta p \approx 1$ bar was chosen, in accordance with [34]. The pressure drop for flowmeters was ignored as it would be negligible with respect of the other quantities.

Ring Distributor and Collector

These ring manifolds function as a hydraulic interface between the cooling jacket and the rest of the system. Both were modeled in a preliminary way as two circular pipes, from which branch out the channels of the jacket (as in a fountain ring manifold). Both the pipes are connected to the main line by a dovetail wye.
The pressure losses were caused by three factors, resulting in: $\Delta p = \Delta p_{friction} + \Delta p_{junction} + \Delta p_{dovetail}$. $\Delta p_{friction}$ represents the losses due to the friction with the walls of the circular pipes, $\Delta p_{junction}$ the losses due to the spillage of part of the flow rate in the channels, and $\Delta p_{dovetail}$ the losses due to the first branch in to the circular pipes of the flow. Due to the complexity to model the real losses of an annular distributor/collector some assumptions were made. The circular pipes were modeled as straight, thanks to the fact that the large relative radius of curvature makes the ratio between the Dean number and the Reynolds number $\frac{De}{Re} \approx 1$, meaning small effects due to curvature. Moreover the mass rate was considered equal and constant in all of the channels. The losses in the circular pipes where therefore computed as in a straight pipe, but considering the mass rate, and so the velocity, to decrease after each encounter with a channel entrance.
A problem emerged form this model was to the fact that the flown incoming in the distributor and exiting in the first channel would lose pressure for 1 bend while the flow outgoing from the last channel would lose pressure from the interaction with all the bends, consequently the pressure drop would be different for each channel. To account for this behavior the losses in every channel were modeled as if they were the first one(the one with the higher velocity), meaning that $\Delta p_{junction} = \frac{1}{2} K_{junction} \rho v_1^2$. The expression for the loss coefficient was given by [30] $K_{junction} = 0.9\left(0.34 + \left(\frac{v_b}{vc}\right)^2\right)$ [2][3] where v_b and v_c are, respectively the fluid velocity in the channel and in the starting pipe. Finally the dovetail pressure loss coefficient was computed from the experimental data in [30] and corresponds to $K_{dovetail} = 0.2$ (assuming a relative curvature of 2).
The collector pressure losses were subdivided in the three same contribution of the distributor. For the channel tee losses the table at page 175 of [30] was used to determine the loss coefficient of $K = -0.5$. Being the density assumed constant in the jacket then there is no reason to assume a different friction loss for the collector than for the distributor. Finally the recombination dovetail wye, interpolating from the curve reported at page 290 of [30], has a loss coefficient of $K = 0.3$.

[2] the formula was considered valid eventhough the flow velocities ratio is bigger than 2
[3] this coefficient refers to the velocity of the common channel

8.12 Water tank sizing

Due to the rapid emptying of the tank, the gas transformation can be considered adiabatic. Consequently:

$$P_i V_i^{\gamma} = P_f V_f^{\gamma} \tag{8.42}$$

and so

$$V_f = V_i \left(\frac{P_i}{P_f} \right)^{\frac{1}{\gamma}} \tag{8.43}$$

where $V_f = V_{N_2} + V_w$ is the total volume of the tank (no water inside the tank after the operations) $V_i = V_{N_2}$ the initial volume of the nitrogen, P_i the initial pressure of the tank, P_f the final pressure and γ is the specific heat ratio, equal to 1.4 for nitrogen. Substituting:

$$V_{N_2} + V_w = V_{N_2} \left(B_d \right)^{\frac{1}{\gamma}} \tag{8.44}$$

The volume of the water V_w is calculated as $V_w = 1.2\frac{\dot{m}(tb+4)}{\rho}$. Where $\dot{m}$ is the mass flow rate, ρ the water density, t_b the total burn time and 1.2 margin to account for residual. Therefore, the volume of the nitrogen is:

$$V_{N_2} = \frac{V_w}{\left((B_d)^{\frac{1}{\gamma}} - 1 \right)} \tag{8.45}$$

and the total tank volume $V_{tank} = V_f = V_{N_2} + V_w$.

The mass of the system is computed by summing together the volume of the cylindrical shell and the hemispherical heads and multiplying the result by the density of the material.

8.13 Pressure vessel sizing

Thanks to the assumption made in 5.2.3 the thicknesses calculation are straight forward. Considering the formulas given in the ASME BVPC manual [25]:

$$t_{\text{shell}} = \frac{P_c \left(\frac{D_{\text{int}}}{2} \right)}{S - 0.6 P_c} \tag{8.46}$$

$$t_{\text{head}} = \frac{P_c \left(\frac{D_{\text{ext}}}{2} \right)}{2S + 0.8 P_c} \tag{8.47}$$

$$t_{\text{divergent}} = \frac{P_t \left(\frac{D_{\text{out}}}{2} \right)}{\cos(\alpha_{\text{div}})(S_n - 0.6 P_t)} \tag{8.48}$$

P_c in the chamber pressure, P_t the throat pressure, S the maximum admissible stress for the aluminum, S_n the maximum admissible stress for the GRCop-84, α_{div} the half-angle of the divergent in correspondence of the throat and D_{int}, D_{ext}, D_{out} the diameters on the inside of the case, the outside of the case and the inside of the exit section of the nozzle respectively.

Calculating the thickness of the convergent while ensuring the same external diameter was more complicated, the formula given by ASME is:

$$t_{convergent,normal} = \frac{P_c \frac{D}{2}}{\cos(\alpha_{conv})(S_n - 0.6P_c)} \tag{8.49}$$

however using $D = D_{int}$ would cause the thickness of the convergent to be extruded outside the case diameter, breaking the constraint smooth outside surface. To fix this issue, the inside diameter of the convergent is written in terms of the outside diameter. Considering :$D_{ext} = D_{con,internal} - 2t_{convergent,radial}$ and expressing $t_{convergent,radial}$ in terms of $t_{convergent,normal}$ to account for the fact that the thickness is extruded normal to the surface and not radial to the nozzle, then:

$$t_{convergent,radial} = \frac{t_{convergent,normal}}{\cos(\alpha_{conv})} \tag{8.50}$$

and substituting in 8.49 yields:

$$t_{convergent,normal} = \frac{P_c \frac{D_{ext}}{2}}{\cos(\alpha_{conv})(S_n - 0.6P_c) + \left(\frac{Pc}{\cos(\alpha_{conv})}\right)}^4 \tag{8.51}$$

The volumes of the components was computed using simple geometry as the difference of the volume of two solids, one bigger (considering the internal radius plus the thickness or the external minus the thickness) and one smaller (considering only the internal radius). This computation of the volume as a difference was performed on the shell, head, convergent and divergent. The mass is then computed by multiplying the volumes of the components by the corresponding densities.

[4]For the computation of the minimum thickess of the convergent in the section characterized by $\frac{A}{A_t} = 2$, the formula 8.46 was used, as the internal diameter was known.

8.14 Bond Coat thermal conductivity

The Bond Coat thermal conductivity was computed as [12]:

$$k(T) = AT^4 + BT^3 + CT^2 + DT + E \tag{8.52}$$

where:

- T = bond coat temperature with a range of: $926 < T < 1223$
- A=$-6.3 * 10^{-11} W/(mK^5)$
- B=$2.0 * 10^{-7} W/(mK^5)$
- C=$-2.0 * 10^{-4} W/(mK^5)$
- D=$1.0 * 10^{-1} W/(mK^5)$

8.15 Simplified panel method for Conjugate Heat Transfer

This appendix details the simplified panel method implemented to resolve the conjugate heat transfer problem. The formulation computes the through-thickness temperature profile of the nozzle wall, the bulk temperature evolution of the coolant, and key thermophysical properties for both the hot combustion gas and the coolant fluid.

Mesh generation The mesh generation routine constructs a two-dimensional structured grid originating from a generic boundary curve $(x, f(x))$. The curve is discretized via uniform sampling along its curvilinear abscissa within the bounded domain $[x_{lb}, x_{up}]$, yielding a set of base nodes $\mathbf{x}_i = [x_i, y_i]^T$. At each node, the local unit normal vector $\mathbf{n}_i$ is computed, adopting a sign convention where $n_y > 0$ denotes the outward direction. The mesh is subsequently built by extruding user-defined layer thicknesses inward and outward from this base geometry. Layer generation proceeds iteratively; for a given layer thickness t_k, the nodal coordinates of the subsequent interface are computed by projecting the current substrate along the local normal field:

$$\mathbf{x}_i^{(k+1)} = \mathbf{x}_i^{(k)} + t_k \mathbf{n}_i$$

Computational panels are defined as the spatial domains bounded between two consecutive extruded normal vectors. Given a sufficiently fine spatial discretization of the smooth nozzle contour, the local longitudinal curvature across an individual panel becomes negligible. Consequently, each discrete segment physically approximates a flat, multi-layered structural boundary. This geometric convergence formally justifies the application of the 1D Planar Composite Wall Approximation. The mesh is subsequently cut in order to obtain a region specified by an upper and lower bound for area ratio, along with a type specifier for each bound (wheter subsonic or supersonic) to generate the requested subMeshes.

Numerical Implementation of the Thermal Solver The thermal solver evaluates the steady-state, one-dimensional conjugate heat transfer across the discretized nozzle panels. The algorithm calculates the through-thickness temperature profile by anchoring the thermal resistance network to a prescribed cold-side boundary condition, T_{cold}, and iteratively resolving the non-linear convective coupling at the hot-gas wall.

Thermochemical Initialization and Gas Dynamics Before resolving the thermal gradients along the wall thickness, the algorithm computes the local thermodynamic and transport properties of the combustion gas. Based on the spatial discretization, local area ratios (A/A_t) are evaluated and passed to a CEA-Based subroutine (implemented with MATLAB based CEA api). This yields the local Mach number (M), Prandtl number (Pr), specific heat (c_p), dynamic viscosity (μ), and static temperature (T_{stat}) for every axial node.

Assuming isentropic expansion, the local total (stagnation) temperature is derived as:

$$T_{\text{tot}} = T_{\text{stat}} \left(1 + \frac{\gamma - 1}{2} M^2\right) \tag{8.53}$$

where γ is the local specific heat ratio. To determine the thermal driving potential at the hot-gas boundary, the adiabatic wall temperature (T_{aw}) is computed via a turbulent recovery ratio formulation. The recovery factor ratio (R), defined relative to the stagnation state, is implemented as:

$$R = \frac{1 + Pr^{1/3} \frac{\gamma-1}{2} M^2}{1 + \frac{\gamma-1}{2} M^2} \tag{8.54}$$

where $Pr^{1/3}$ represents the standard turbulent recovery factor for the boundary layer. The local adiabatic wall temperature is subsequently evaluated directly from the total temperature:

$$T_{\text{aw}} = R\, T_{\text{tot}} \tag{8.55}$$

Governing Equations of the Thermal Circuit For each discrete panel, the transmural heat flux ($\dot{q}''$) is governed by the total thermal resistance opposing the potential difference between the adiabatic wall and the enforced cold-wall boundary. The total conductive resistance of the composite wall (R''_{cond}) is treated as a geometric and material constant:

$$R''_{\text{cond}} = \sum_{k=1}^{N_{\text{layers}}} \frac{t_k}{k_k} \tag{8.56}$$

where t_k and k_k represent the thickness and thermal conductivity of the k-th layer, respectively.

The fundamental non-linearity of the system arises from the convective heat transfer coefficient (h_{conv}). Modeled via the Bartz correlation (already presented in 8.8), h_{conv} relies on a boundary layer property correction factor σ (already presented in 8.8) to account for the steep thermal gradients between the freestream gas and the wall. Because σ is a strong function of the hot-gas wall temperature (T_{wg}), the convective coefficient becomes dynamically coupled to the unknown wall state: $h_{\text{conv}} = f(T_{\text{wg}})$.

Consequently, the global steady-state heat flux must satisfy the following relation:

$$\dot{q}'' = \frac{T_{\text{aw}} - T_{\text{cold}}}{\frac{1}{h_{\text{conv}}(T_{\text{wg}})} + R''_{\text{cond}}} \tag{8.57}$$

Iterative Root-Finding Algorithm Because T_{wg} is required to compute h_{conv}, yet h_{conv} dictates the heat flux required to determine T_{wg}, the system cannot be solved analytically. Instead, an iterative zero-crossing algorithm is deployed to enforce thermodynamic equilibrium at each spatial node.

The algorithmic logic proceeds through the following steps:

1. **Initialization:** An initial guess for the hot-gas wall temperature ($T_{\text{wg,guess}}$) is established as a fraction of the local stagnation temperature.

2. **Convective Evaluation:** The Bartz equation evaluates h_{conv} utilizing the current $T_{\text{wg,guess}}$ to correctly apply the σ correction factor.

3. **Flux Computation:** The instantaneous 1D heat flux $\dot{q}''$ is calculated via Equation 8.57.

4. **Thermal Integration:** The through-thickness temperature profile is integrated moving inward from the known,outer cold boundary condition (T_{cold}). Following Fourier's law, the temperature drop across each discrete layer is strictly linear:

$$T_{k+1} = T_k + \dot{q}'' \left(\frac{t_k}{k_k} \right) \tag{8.58}$$

5. **Residual Evaluation:** Once the integration reaches the innermost layer, a calculated hot-gas wall temperature ($T_{\text{wg,calc}}$) is obtained. The solver evaluates the thermal residual:

$$\mathcal{R}(T_{\text{wg}}) = T_{\text{wg,calc}} - T_{\text{wg,guess}} \tag{8.59}$$

The root-finding scheme iteratively updates $T_{\text{wg,guess}}$ until the residual $\mathcal{R}(T_{\text{wg}})$ is driven to zero within a strict numerical tolerance. Upon convergence, the exact steady-state heat flux, the physically consistent convective coefficient, and the complete through-thickness temperature array are extracted before the solver advances to the subsequent axial node and repeats the process.

Upon completing the spatial march along the entire nozzle contour, the solver outputs a comprehensive discrete thermal and fluid dynamic dataset. Specifically, the function returns a spatially resolved, two-dimensional temperature matrix that captures the exact thermal state at every structural interface (from the inner hot side wall temperature to the outer, imposed cold temperature) for each evaluated panel. In addition to this through-thickness thermal mapping, the algorithm provides discrete arrays of the steady-state wall heat flux ($\dot{q}''$) and the coupled convective heat transfer coefficient (h_{conv}), resolved exclusively at the panel midpoints. To support further system analysis, the output structure also aggregates vectors of the localized thermochemical flow properties—such as the Mach number, specific heat capacity, dynamic viscosity, and Prandtl number—evaluated at each discretized cross-section.

Cooling Jacket Solver The discrete geometrical mesh and the converged through-wall heat flux arrays (where heat flux is defined for evey panel) establish the thermal loading conditions for the cooling jacket. By coupling these spatially prescribed heat fluxes with the initial thermodynamic boundary conditions of the coolant fluid, this secondary solver executes the structural sizing and fluid-dynamic resolution of the cooling channel architecture.

Coolant Energy Balance and Thermodynamic State The cooling jacket is modeled as a parallel array of N_{channels} rectangular conduits. The solver iterates from the coolant inlet to the outlet in a 1D marching scheme. For each discrete panel segment i, the algorithm first updates the bulk fluid temperature via a steady-state energy balance. Because the bulk fluid temperature is assumed to be uniform across all channels at any given axial station, the energy balance is evaluated globally for the entire nozzle cross-section.
Assuming the wall-normal heat flux ($\dot{q}_i''$) evaluated by the hot-wall solver is completely absorbed by the coolant, the fluid temperature exiting the segment ($T_{\text{eval},i}$) is computed as:

$$T_{\text{eval},i} = T_{\text{in},i} + \frac{\dot{q}_i'' S_i}{\dot{m} c_p(T_{\text{in},i})} \tag{8.60}$$

where $T_{\text{in},i}$ is the segment inlet temperature, S_i is the conical surface area of the local nozzle wall segment, $\dot{m}$ is the total coolant mass flow rate (imposed by the user) through the entire jacket, and c_p is the temperature-dependent specific heat of the coolant (water). With the bulk fluid temperature resolved, the algorithm computes the target convective heat transfer coefficient (h_{req}) required to maintain the prescribed cold-wall boundary temperature (T_{panel}) under the imposed thermal load:

$$h_{\text{req},i} = \frac{\dot{q}_i''}{T_{\text{panel},i} - T_{\text{eval},i}} \tag{8.61}$$

T_{Panel} is a known quantity, being the temperature of the outer wall of the mesh which is imposed by the user (equal to the T_{cold} for the wall solver);

Geometric Constraints and Heat Transfer Coupling To geometrically size the channel, the solver imposes a structural boundary condition: the local channel width (a_i) is constrained by the local nozzle circumference and the total number of channels:

$$a_i = \frac{2\pi r_i}{N_{\text{channels}}} \tag{8.62}$$

where r_i is the local nozzle radius. Consequently, the local channel height (b_i), the fluid velocity (v_i), and the effective hydraulic diameter ($D_{h,i}$) remain strictly coupled unknown variables.
To achieve the required heat transfer coefficient $h_{\text{req},i}$, the fluid dynamics must satisfy a selected empirical Nusselt number correlation (either Dittus-Boelter or Gnielinski). The analytical required Nusselt number (Nu_{req}) is defined as:

$$Nu_{\text{req}} = \frac{h_{\text{req},i} D_{h,i}}{k(T_{\text{eval},i})} \tag{8.63}$$

where k is the temperature-dependent thermal conductivity of the coolant.

Non-Linear System Resolution Prior to initiating the numerical solver, the requisite thermophysical properties of the coolant are evaluated at the updated bulk fluid temperature ($T_{\text{eval},i}$). Assuming liquid water as the working fluid, the algorithm computes the temperature-dependent dynamic viscosity (μ), thermal conductivity (k), and specific heat capacity (c_p) utilizing high-fidelity empirical polynomial fits (see Appendix 8.9). For this incompressible regime, the fluid density (ρ) is modeled as a constant.

To enforce mass continuity, convective heat transfer equivalency, and geometric validity simultaneously, a multivariable root-finding algorithm (`fsolve`) is deployed. The unconstrained computational state vector $\mathbf{y} = [y_1, y_2, y_3]^T$ maps to the physical variables via an exponential transformation ($v = \exp(y_1)$, $D_h = \exp(y_2)$, $b = \exp(y_3)$) to strictly prevent the solver from exploring non-physical negative quantities.
The numerical solver drives the following system of normalized residuals to zero:

$$\begin{bmatrix} \mathcal{R}_1 \\ \mathcal{R}_2 \\ \mathcal{R}_3 \end{bmatrix} = \begin{bmatrix} 1 - \frac{N_{\text{channels}} \rho a_i b_i v_i}{\dot{m}} \\ 1 - \frac{N u_{\text{req}}}{N u_{\text{corr}}(Re, Pr)} \\ 1 - \frac{\frac{2 a_i b_i}{a_i + b_i}}{D_{h,i}} \end{bmatrix} = \mathbf{0} \tag{8.64}$$

- $\mathcal{R}_1$ **(Mass Continuity):** Ensures the total mass flow rate is perfectly conserved across the integrated cross-sectional area of all parallel channels.
- $\mathcal{R}_2$ **(Thermal Coupling):** Enforces equality between the analytically required Nusselt number and the empirical correlation Nusselt number (Nu_{corr}), effectively sizing the flow to provide the exact cooling capacity needed.
- $\mathcal{R}_3$ **(Geometric Definition):** Constrains the unknown hydraulic diameter to strictly match the analytical formulation for a rectangular cross-section.

Upon driving the residual vector to within numerical tolerance, the solver extracts the converged channel height, fluid velocity, hydraulic diameter, fluid velocity and other quantities. Furthermore, the local Reynolds number (Re) and Prandtl number (Pr) are evaluated and recorded; this explicitly allows for post-processing verification to ensure the flow regime remains strictly within the mathematical validity bounds of the selected empirical Nusselt correlation. Finally, the segment exit temperature is assigned as the inlet temperature for the adjacent downstream segment ($T_{\text{in},i+1} = T_{\text{eval},i}$), and the spatial marching scheme proceeds.